RESIDENCE 5105

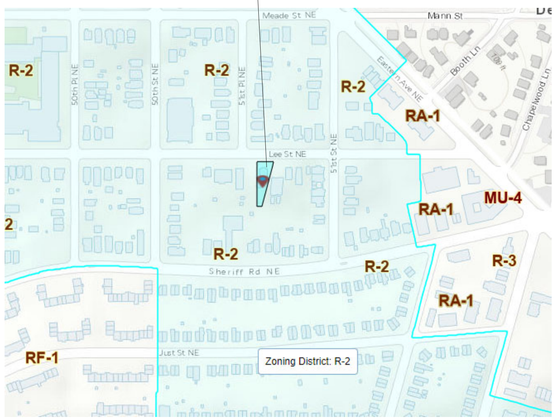
5105 LEE ST NE
WASHINGTON DC 20019

LOT: 0027 SQUARE: 5200

SCOPE OF WORK

ALTERATION AND REPAIR OF AN EXISTING DETACHED SINGLE FAMILY CELLAR AND 2 STORY DETACHED SFD. TO DESIGN AND FINISH INTRIOR OF EXISTING CELLAR. TO FINISH MAIN AND UPPER LEVELS. TO DESIGN NEW MEP THROUGH OUT.

LOCATION



VICINITY MAP





OWNER INFORMATION

SELECTA INVESTMENT GROUP LLC

OWNER ADDRESS

5105 LEE ST NE
WASHINGTON DC 20019-4043

PROJECT TEAM:

DESIGN ARCHITECT:
ONE DESIGN
DAGMAWI GEBREKIDAN
(571.225.7211)
(dagi.abebe@gmail.com)
ANTHONY UDOKA
LIAH MEBRAHTU

MEP:
KHALID LAMAFI

DRAWING INDEX

Sheet List

Sheet Number	Sheet Name	Current Revision
000	COVER SHEET	
A001	EXISTING FLOOR PLANS	
A100	PROPOSED FLOOR PLANS	
A101	SECTION & WALL DETAILS	
A201	ELEVATIONS, WINDOW & DOOR SCH.	
A301	MATERIALS SCH.	
A401	GENERAL INSULATION NOTES	
A402	FIRE RELATED GENERAL DETAIL	
E100	ELECTRICAL COVER SHEET	
E200	ELECTRICAL LIGHTING FLOOR PLANS	
E300	ELECTRICAL POWER FLOOR PLANS	
E400	ELECTRICAL PANEL SCHEDULES & RISER	
M100	MECHANICAL COVER SHEET & SCHEDULES	
M200	MECHANICAL FLOOR PLAN	
M300	MECHANICAL DETAILS	
P100	PLUMBING COVER SHEET	
P200	PLUMBING DETAILS	
P300	PLUMBING SANITARY FLOOR PLANS	
P400	PLUMBING DOMESTIC WATER FLOOR PLANS	
P500	PLUMBING RISER DIAGRAMS	

Sheet List

Sheet Number	Sheet Name	Current Revision

SYMBOLS

- PLAN DETAIL REFERENCE (X / XX: DETAIL REFERENCE / SHEET NUMBER)
- ELEVATION REFERENCE (X / XX: DETAIL REFERENCE / SHEET NUMBER)
- SECTION REFERENCE (X / XX: DETAIL REFERENCE / SHEET NUMBER)
- EXISTING WALL TO BE REMOVED
- EXISTING WALL TO REMAIN
- NEW BRICK WALL
- NEW CMU WALL
- NEW STUD WALL
- DOOR DESIGNATION
- WINDOW DESIGNATION
- PARTITION TYPE
- KEY NOTE
- XXXX ROOM NUMBER
- X FIXTURE TYPE
- ELEVATION MARKER
- X XX INTERIOR ELEVATION DESIGNATION

ABBREVIATIONS

AFF	ABOVE FINISH FLOOR	MAT	MATERIAL
AA	ALL AROUND	MAX	MAXIMUM
ACOUS	ACOUSTICAL	MC	MILLWORK CONTRACTOR
ACT	ACOUSTICAL CEILING TILE	MDF	MEDIUM DENSITY FIBERBOARD
ADJ.	ADJUSTABLE	MDO	MEDIUM DENSITY OVERLAY
ALUM	ALUMINUM	MECH	MECHANICAL
ANOD	ANODIZED	MIN.	MINIMUM
BD	BOARD	MISC.	MISCELLANEOUS
BLKG	BLOCKING	MLDG	MOLDING
BS	BOTH SIDES	MO	MASONRY OPENING
<	CENTER LINE	MTD	MOUNTED
CLG	CEILING	MTL	METAL
CMU	CONCRETE MASONRY UNIT	OC	ON CENTER
CONT	CONTINUOUS	OPP	OPPOSITE
DIA	DIAMETER	PLYWD	PLYWOOD
DIM	DIMENSION	PL	PLATE
DN	DOWN	POL	POLISHED
DOUG	DOUGLAS	PTD	PAINTED
DR	DOOR	RCP	REFLECTED CEILING PLAN
DTL	DETAIL	RND	ROUND
DWG	DRAWING	RO	ROUGH OPENING
ELEC	ELECTRICAL	SC	SOLID CORE
ELEV	ELEVATION	SIM	SIMILAR
EQ	EQUAL	SHT	SHEET
EQUIP	EQUIPMENT	STND	STAINED
EXST.	EXISTING	STL	STEEL
FF	FINISH FLOOR	STOR	STORAGE
FIXT	FIXTURE	TEL	TELEPHONE
FLR	FLOOR	TYP	TYPICAL
FLUOR	FLUORESCENT	TME	TO MATCH EXISTING
GFI	GROUND FAULT INTERRUPTER	UON	UNLESS OTHERWISE NOTED
GC	GENERAL CONTRACTOR	VCT	VINYL COMPOSITION TILE
GL	GLAZING	VERT	VERTICAL
GYP	GYPSUM	VEN	VENEER
GWB	GYPSUM WALL BOARD	VIF	VERIFY IN FIELD
HDWR	HARDWARE	VWC	VINYL WALL COVERING
HM	HOLLOW METAL	W/	WITH

GENERAL NOTES

DIVISION 1- GENERAL REQUIREMENTS

1. THE SUB-CONTRACTOR SHALL READ ALL GENERAL AND SPECIFIC NOTES AND BE BOUND TO THEIR REQUIREMENTS SUB-CONTRACTOR SHALL VISIT SITE AND EXAMINE EXISTING CONDITIONS PRIOR TO BID.
2. THE SUB-CONTRACTOR SHALL VISIT THE SITE AND FAMILIARIZE HIMSELF WITH THE CONDITIONS UNDER WHICH THE WORK IS TO BE PREFORMED.
3. THE BID SHALL BE BASED ON THE USE OF ALL EQUIPMENT, MATERIALS, MANUFACTURERS AND SYSTEMS DESCRIBED IN THIS SET OF CONSTRUCTION CONTACT DOCUMENTS UNLESS SPECIFICALLY INDICATED IN THE PRICE SUBMISSION. UNLESS AN ITEM IS SPECIFICALLY EXCLUDED OR A SUBSTITUTION IS LISTED ON THE SUBMISSION BY THE SUB-CONTRACTOR, HE SHALL BE RESPONSIBLE FOR THE FULL EXTENT OF ALL WORK CONTAINED IN THE CONTRACT DOCUMENTS.
4. ANY QUESTIONS OR CONFLICTS REGARDING ARCHITECTURAL DRAWINGS OR SPECIFICATIONS ARE TO BE IN WRITING AND DIRECTED TO RICH MARKUS AND RICH MARKUS ARCHITECTS.
5. IF CONFLICTS ARE DISCOVERED AT ANY TIME BETWEEN ENGINEERING AND ARCHITECTURAL DRAWINGS, ARCHITECT SHALL BE CONTACTED AT ONCE TO RESOLVE THE PROBLEM. IF A ARCHITECT IS NOT CONTACTED, SUB CONTRACTOR SHALL ACCEPTANT RESPONSIBILITY TO CORRECT ANY WORK INSTALLED THAT IS NOT IN ACCORDANCE WITH THE DWGS.
6. IF CONFLICTS ARE DISCOVERED AT ANY TIME BETWEEN SITE CONDITIONS AND ARCHITECTURAL DWGS, ARCHITECT SHALL EXACT LOCATIONS OF ALL UTILITIES PRIOR TO CUTTING AND DIGGING. ALL SLAB PENETRATION SHALL BE CONTACTED AT ONCE TO RESOLVE ANY DISCREPANCIES, OTHERWISE CONTRACTOR TAKES FULL RESPONSIBILITY.
7. IN CONFLICTS BETWEEN ARCHITECTURAL AND ENGINEERING SPECIFICATIONS OF FINISHES, ARCHITECTURAL DWGS SHALL TAKE PRECEDENCE AND CONFLICTS SHOULD BE BROUGHT TO THE ATTENTION OF THE ARCHITECT IMMEDIATELY.
8. SUB- CONTRACTOR SHALL VERIFY ALL DIMENSIONS AND QUANTITIES IN THE FIELD
9. WHERE SLAB CUTTING IS REQUIRED, IT SHALL BE SUB-CONTRACTOR'S RESPONSIBILITY TO VERIFY EXACT LOCATIONS OF ALL UTILITIES PRIOR TO CUTTING AND DIGGING. ALL SLAB PENETRATION SHALL BE DONE BY EITHER CORE DRILLING OR SAW CUTTING AND DIGGING. ALL SLAB ON GRADE, SUB-CONTRACTOR SHALL BACKFILL IN LIFTS NOT TO EXCEED 8" AND TAMPER USING A MECHANICAL COMPACTOR. NEW PATCHING SHALL BE THE SAME THICKNESS AS EXISTING SLAB AND, SHALL ALIGN AND MAINTAIN THE LEVEL OF EXISTING SLAB.
10. SUB-CONTRACTOR SHALL PERFORM ALL WORK IN ACCORDANCE WITH NATIONAL, STATE AND LOCAL CODES.
11. ALL PERMITS, INSPECTIONS AND FEES ARE TO BE OBTAINED, SECURED AND PAID FOR BY THE SUB-CONTRACTOR
12. SUB-CONTRACTOR MAY SUBSTITUTE MATERIALS AND/OR CONSTRUCTION ITEMS AND METHODS UPON SUBMITTAL OF A SUBSTITUTION IN WRITHING OR BY SAMPLE TO ARCHITECT. ANY COST CHANGES MUST ALSO BE SUBMITTED. THE ARCHITECT WILL COMMUNICATE THE DECISION TO THE SUB-CONTRACTOR BY LETTER IF THE SUBMISSION IS ACCEPTED. SUBSTITUTION FOR LONG LEAD ITEMS MUST BE PROPOSED EARLY ENOUGH IN THE CONSTRUCTION PROCESSES THAT NO DELAY IS CREATED IN COMPLETION IF THE PROJECT DUE TO THE SUBSTITUTION, OTHERWISE THE SUB CONTRACTOR SHALL ASSUME RESPONSIBILITY FOR ANY COST TO THE OWNER ARISING OUT OF SUCH WORK.
13. WARRANTIES, GUARANTEES AND MANUFACTURER'S INSTRUCTIONS ON EQUIPMENT FURNISHED AND INSTALLED BY THE SUB-CONTRACTOR SHALL BE ASSEMBLED BY NOTEBOOK GIVEN TO THE OWNER AT THE TIME OF OCCUPANCY.
14. AT COMPLETION OF THE WORK, THE SUB-CONTRACTOR SHALL ENSURE ALL SURFACES ARE CLEAN AND UNMARKED, AND THAT ALL SYSTEMS ARE OPERATING PROPERLY.
15. SUB-CONTRACTOR SHALL REPAIR ALL DAMAGE CAUSED BY HIM OR HIS PERSONNEL
16. SUB-CONTRACTOR SHALL CLEAN ALL WINDOWS AT THE COMPLETION OF CONSTRUCTION.
17. SUB-CONTRACTOR SHALL ENSURE THAT ALL PAINT HAS BEEN MOVED FROM ALL SURFACES,
18. ALL MISCELLANEOUS NAILS, HANGERS, STAPLES, WIRES AND WIRE CONDUITS SHALL BE REMOVED FROM WALLS AND CEILINGS.
19. HVAC SUB CONTRACTOR TO CLEAN AND CHECK EXISTING HVAC SYSTEM AND REPORT TO ARCHITECT IF ANY PROBLEMS OR REPAIRS NEEDED.`
20. SUB-CONTRACTOR SHALL EXAMINE ALL DRAWINGS RELATED TO HIS AND OTHERS TRADES, AND SHALL BE FULLY INFORMED AS TO THE EXTENT OF THIS CONTRACT AND ALL INCLUDING WORK ON PLANS.
21. ALL SYSTEMS AND EQUIPMENT TO BE GUARANTEED FOR ONE YEAR FREE LABOR AND MATERIALS.

GENERAL NOTES / CODE ANALYSIS

GENERAL INFORMATION		
USE GROUP	EXISTING LOT	PROPOSED BUILDING
TYPE OF CONSTRUCTION	VB	VB
FULLY SUPPRESSED (SPRINKLED)	-	NO
LOT AREA	7826	7826
BUILDING FOOTPRINT	1299	1299
TOTAL SITE SF		TOTAL BUILDING SF
BUILDING FOOTPRINT	1299	ABOVE GRADE: 1299 SF (2)
PROPOSED GREEN AREA	N/A	CELLAR : 1299 SF
PROPOSED PERMEABLE PAVEMENT	N/A	TOTAL = 3,897 SF
BUILDING HEIGHT	2 story +CELLAR	24' -0" / 2 STORIES + CELLAR
NUMBER OF EXITS	3	3
SOUND TRANSMISSION	-	STC50 IIC50
FIRE ALARM:	-	YES; HARDWIRED AND INTERCONNECTED ON DEDICATED
SMOKE DETECTORS		CIRCUIT WITH BATTERY BACKUP
INSULATION:		YES; EXTERIOR WALLS @ R19; ROOF @ R-49

COVERING CODES:	
BUILDING	2015 IBC AND 2017 DCMR12A SUPPLEMENT 2015 IEBC
COVERING CODES:	2017 DISTRICT OF COLOMBIA BUILDING CODE
MECHANICAL	2017 DISTRICT OF COLUMBIA MECHANICAL CODE
ELECTRICAL	2014 INTERNATIONAL NFPA-70
PLUMBING	2017 DISTRICT OF COLUMBIA PLUMBING CODE
GREEN	2017 DISTRICT OF COLUMBIA FIRE CODE
FIRE	2017 DISTRICT OF COLUMBIA GREEN CONSTRUCTION CODE
ENERGY	2017 DISTRICT OF COLUMBIA ENERGY CONSERVATION CODE

FIRE RATINGS		
FIRE SEPARATION WALLS (PROPERTY LINE WALLS)	2 HR:	(UL U301 - above grade / U 905 - below grade)
EXTERIOR BEARING WALLS:	1 HR:	(UL U301 - above grade / U 905 - below grade)
STAIR ENCLOSURES	1 HR:	(UL U305)
COLUMNS AND GIRDERS	(none provided)	-
FLOOR / CEILING CONSTRUCTION	1 HR:	(UL L 501)
ROOF CONSTRUCTION	1 HR:	(UL L 501)
TENANT SEPARATION / CORRIDOR	1 HR:	(UL L 305)

GENERAL NOTES / ZONING CODE ANALYSIS

5105 LEE ST NE
WASHINGTON DC 20019-4043

ZONING DISTRICT	R-2	TOTAL LOT AREA	7826 SF
USE GROUP	R-3	TOTAL LOT WIDTH	62'-0" -FRONT & 165'-0" BACK
LOT	0027		
SQUARE	5200		

REQUIREMENT	EXISTING	ALLOWABLE / REQUIRED	PROPOSED
LOT OCCUPANCY	20%	60% (4,695.6 SF)	20%
HEIGHT	25 FT	40 FT	25 FT
F.A.R.	N/A	N/A	N/A
SIDEYARD WIDTH	N/A	8FT	N/A
REAR YARD SETBACK	N/A	20 FT MIN	N/A
PARKING	N/A	2	N/A
FRONT YARD	N/A	NO REQUIREMENT	N/A
DWELLING UNITS	N/A	SFD	N/A

5105 LEE ST NE

ORGANIZATION FOR NEW ENVIRONMENTS LLC

dagi.abebe@gmail.com
Tel. 571.225.7211

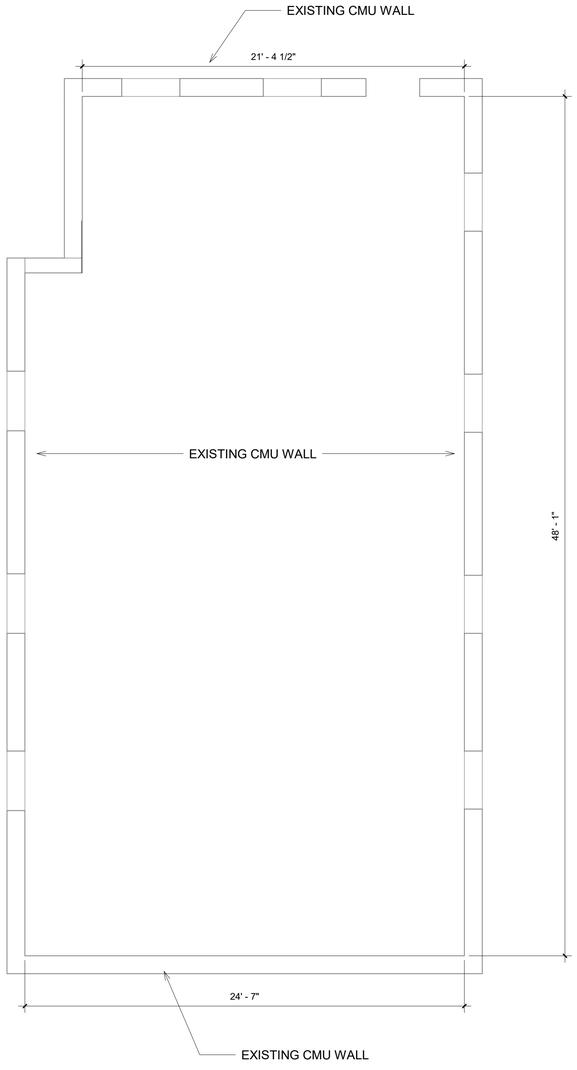
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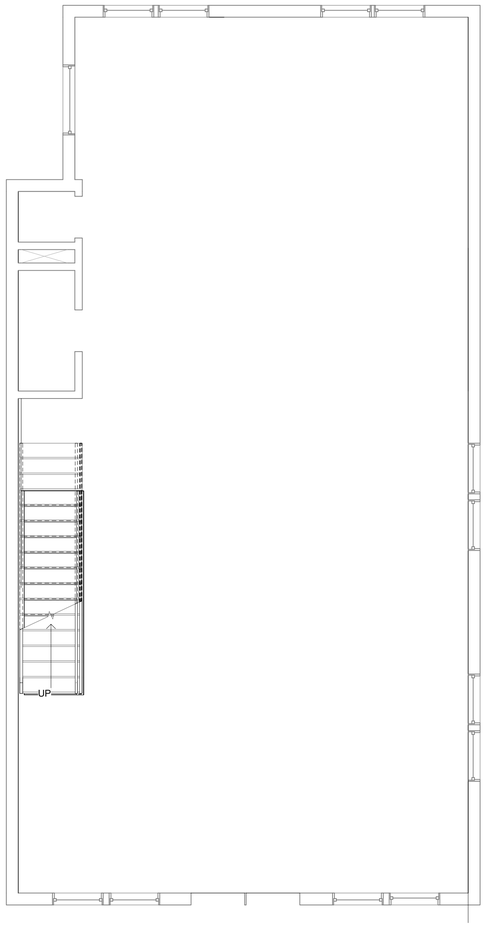
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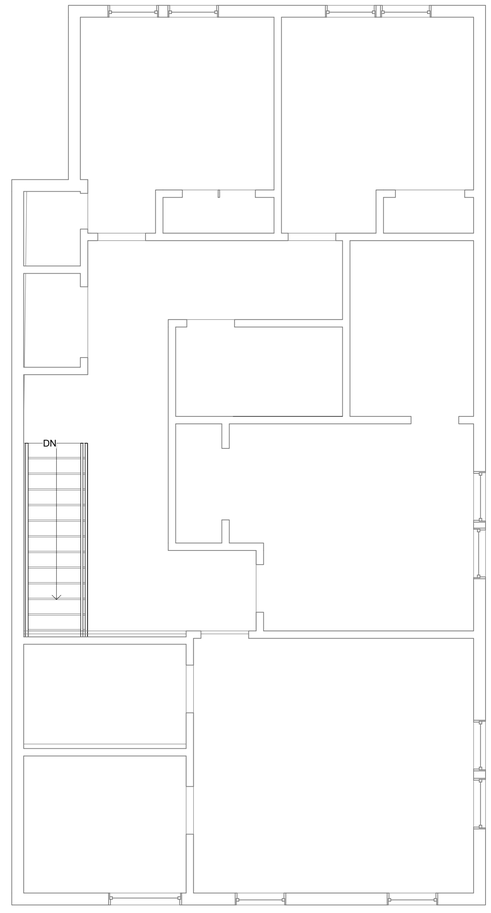
000



1 A001 EXISTING - CELLAR FLOOR PLAN
1/4" = 1'-0"



2 A001 EXISTING - FIRST FLOOR PLAN
1/4" = 1'-0"



3 A001 EXISTING - SECOND FLOOR PLAN
1/4" = 1'-0"

5105 LEE ST NE

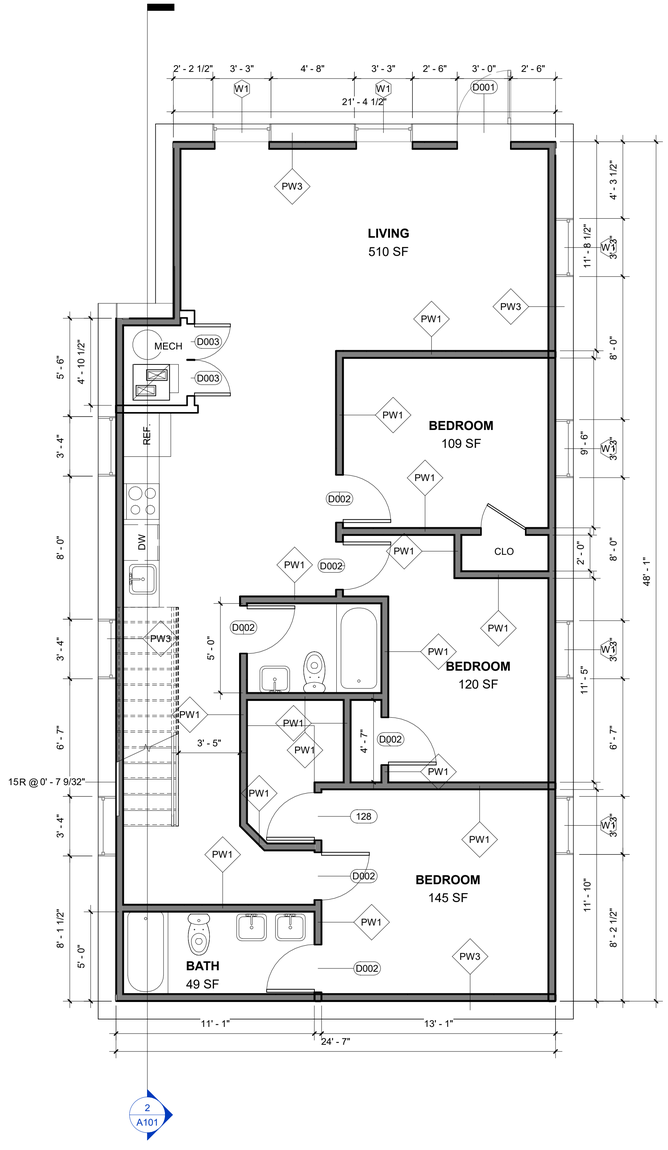
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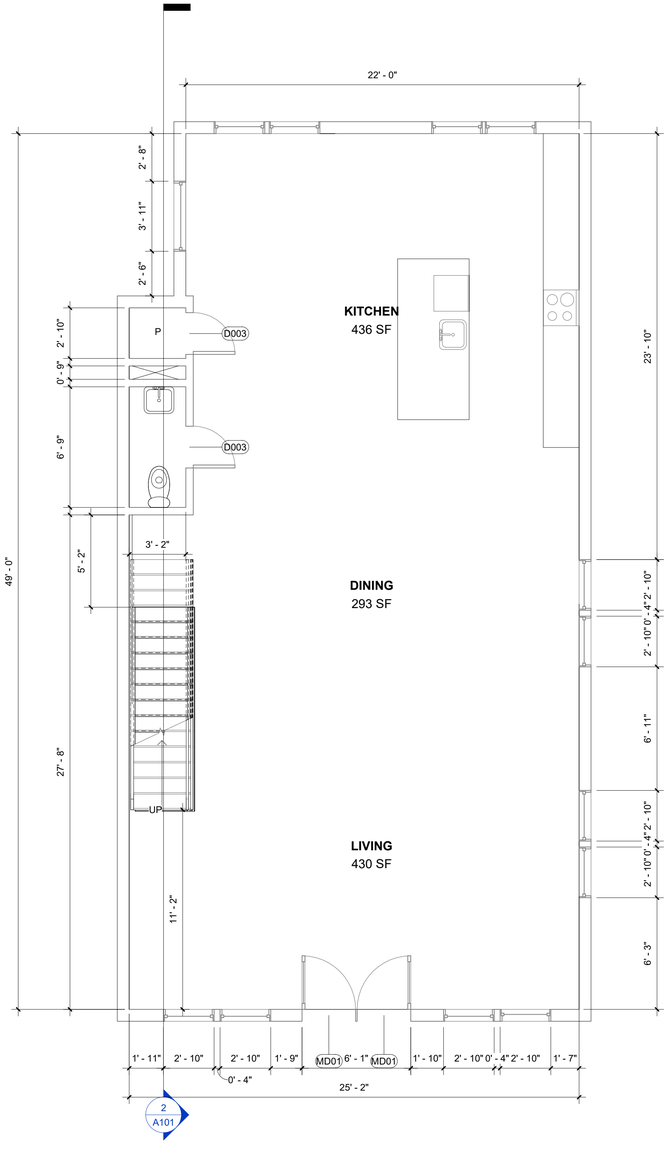
EXISTING FLOOR PLANS

A001



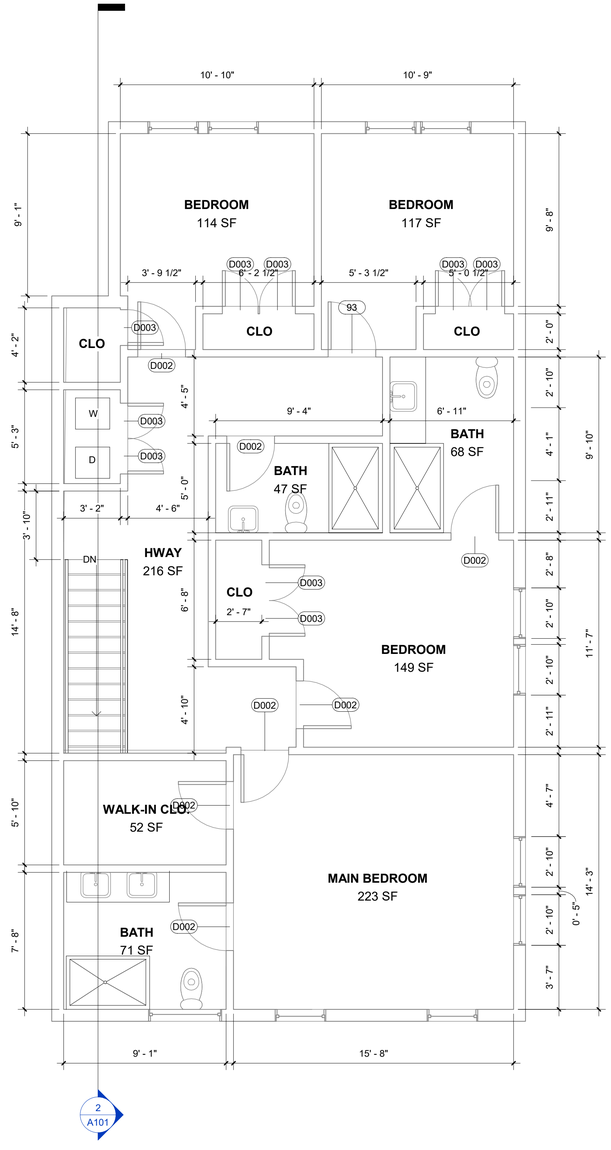
2
A101

1 A100 PROPOSED - CELLAR FLOOR PLAN
1/4" = 1'-0"



2
A101

2 A100 PROPOSED - 1ST FLOOR
1/4" = 1'-0"



2
A101

3 A100 PROPOSED - 2ND FLOOR
1/4" = 1'-0"

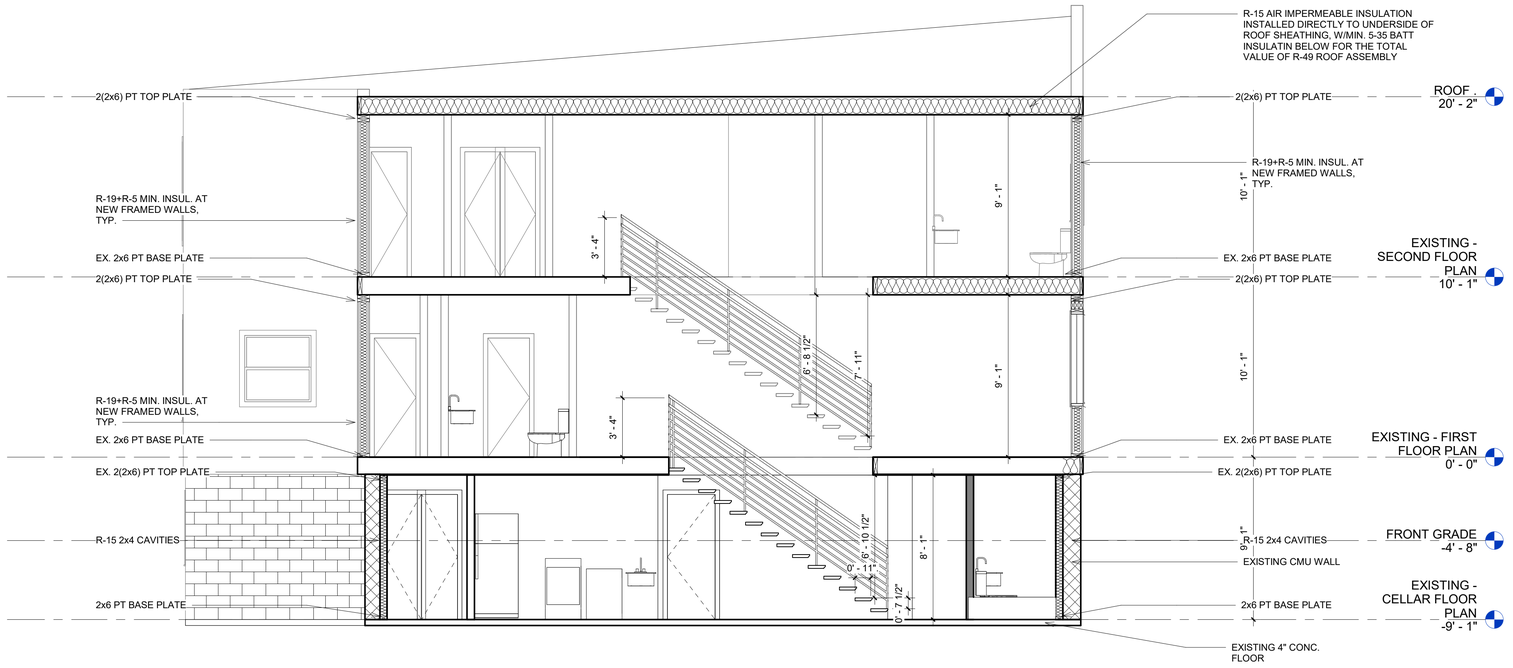
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ORGANIZATION FOR NEW ENVIRONMENTS LLC
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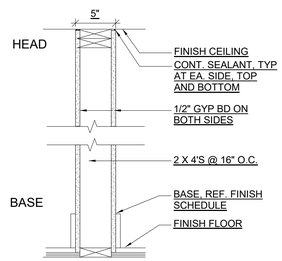
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PROPOSED FLOOR PLANS

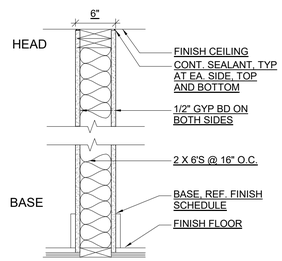
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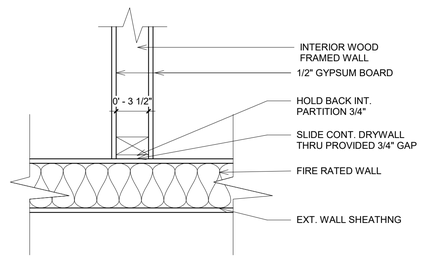
2 A101 **Section 1**
1/4" = 1'-0"

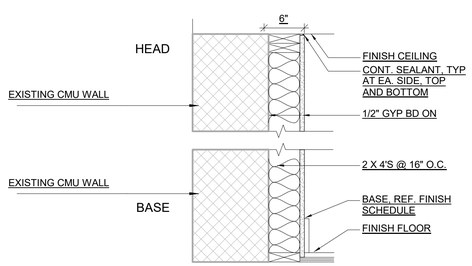


PW1 - INTERIOR PARTITION DETAIL



PW2 - EXTERIOR WALL DETAIL





PW3 - EXTERIOR WITH CMU WALL DETAIL

3 A101 **PARTITION TYPES**
1 1/2" = 1'-0"

4 A101 **CONNECTION DETAIL**
1 1/2" = 1'-0"

LEGEND:

EXISTING CMU WALL	
EXISTING WALL	
DEMO WALL	
NEW WALL	

N

5105 LEE ST NE

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dagi.abebe@gmail.com
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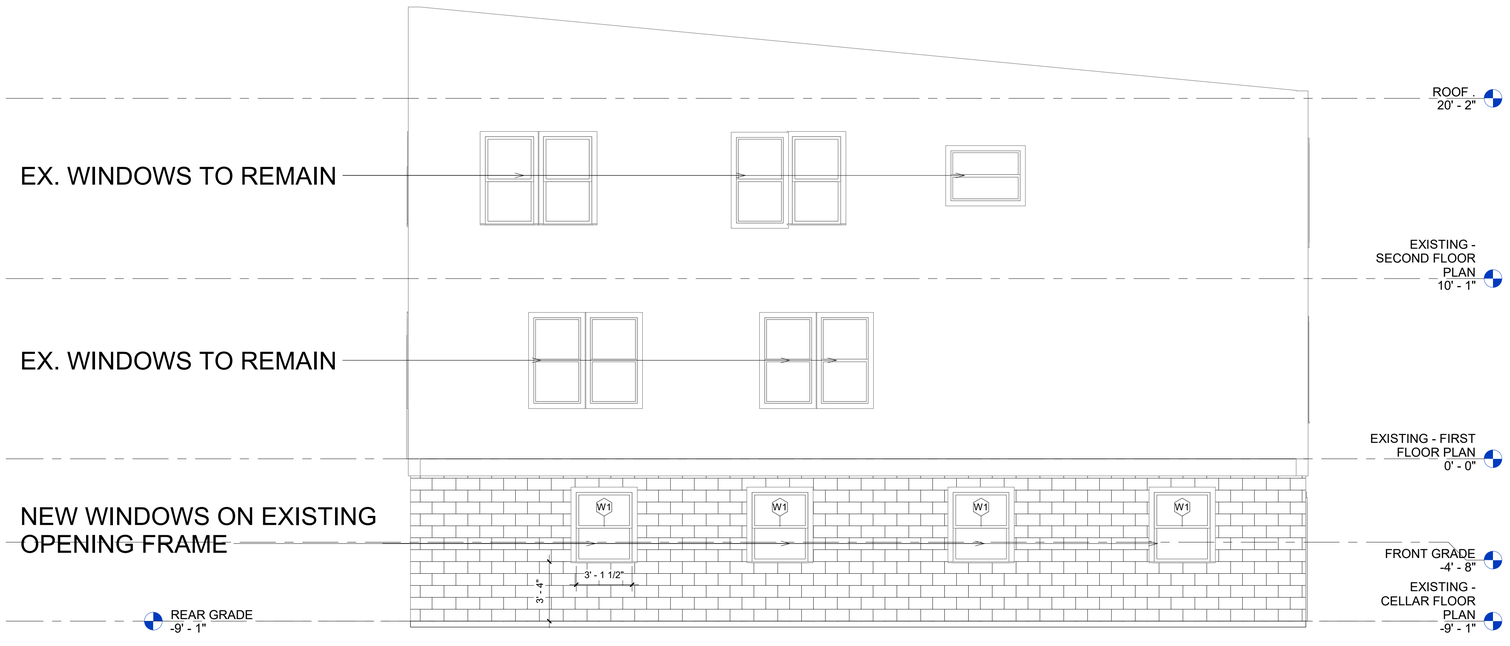
REVISION #	
SCALE	AS INDICATED
ISSUE DATE	JUNE 2020

SECTION & WALL DETAILS

A101



1 A201 EXISTING REAR ELEVATIONS
1/4" = 1'-0"



2 A201 RIGHT SIDE ELEVATION
1/4" = 1'-0"

WINDOW NAME	W1
WINDOW PLAN	
WINDOW FRONT ELEVATION	3' - 1 1/2" · 9' - 0" · 3' - 4"

Fenestration U-factor and SHGC must be less than 0.30 and 0.40 respectively. Windows, skylights, and sliding glass doors shall have an air infiltration rate of no more than 0.3 cfm per square foot, and swinging doors no more than 0.5 cfm per square foot.

DOOR NAME	D001	D002	D003
DOOR PLAN			
DOOR FRONT ELEVATION	3' - 0" · 7' -0"	2' - 8" · 7' - 0"	2' - 0" · 7' - 0"

WINDOW & DOOR SCHEDULE
1/4" = 1'-0"

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SCALE	AS INDICATED
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ELEVATIONS, WINDOW & DOOR SCH.

A201

INTERIOR FINISH SCHEDULE - WHERE APPLICABLE (SEE DRAWINGS)

Designation	Description	Manufacturer	Finish/Style/Color/Size	Notes
P-1	PAINT	BENJAMIN MOORE	BALBOA MIST 1549, FLAT	THROUGHOUT WHOLE BUILDING
P-2	PAINT	TBD	TBD	
P-3	PAINT	TBD	TBD	
WD-1	WOOD FLOOR		3 COATS OF OIL BASE, STAIN TBD	4" SAND IN PLACE RED OAK #1 COMMON W/
CT-1	CERAMIC TILE		MELTED ICE, SIZE 12X24	MASTER BATH FLOOR TILE
CT-2	CERAMIC TILE		MELTED ICE, SIZE 12X24	MASTER BATH WALL TILE
CT-3	CERAMIC TILE		WOX HEX, SIZE 1X1, ONYX CLOUD	MASTER BATH SHOWER FLOOR TILE
CT-4	CERAMIC TILE		TIRAMISU GLASS STIX	MASTER BATH NICHE
CT-5	CERAMIC TILE		MATRIX BRIGHT WHITE, SIZE 12X24	GUEST BATH FLOOR TILE
CT-6	CERAMIC TILE		GLOSS BRIGHT WHITE, SIZE 4X10, STACKED	GUEST BATH WALL TILE
CT-7	CERAMIC TILE		PENNY ROUNDS WHITE GLOSS	GUEST BATH NICHE
CT-8	CERAMIC TILE		'3X6 WHITE GLOSS '3X9 WHITE GLASS' SUBWAY TILE	KITCHEN BACKSPLASH
CNT-1	COUNTERTOP		SUPER WHITE QUARTZITE 3CM	KITCHEN
CNT-2	COUNTERTOP		CARERRA MARBLE 3CM	MASTER AND GUEST BATH

(TBS BY OWNER)

HARDWARE SCHEDULE

Hardware Set	Qty	Item	Manufacturer	Manufacturer #	Finish	Function	Notes
A BEDROOM AND BATHROOM DOOR	1	LOCKSET	SCHLAGE S-SERIES	JUPITER	626	PRIVACY	
	1	STRIKE	SCHLAGE S-SERIES	10-025	626		
	3	HINGE	McKINNEY	TA2714	US26D		
	1	WALLSTOP	IVES	WS447	US26D		
		SILENCER	GLYNN JOHNSON	GJ64			
B CLOSET DOOR	1	LOCKSET	SCHLAGE S-SERIES	JUPITER	626	PASSAGE	
	1	STRIKE	SCHLAGE S-SERIES	10-025	626		
	3	HINGE	McKINNEY	TA2714	US26D		
		SILENCER	GLYNN JOHNSON	GJ64			
C DOUBLE CLOSET DOORS	1	LOCKSET	SCHLAGE S-SERIES	JUPITER	626	SEMI-PRIVACY	
	1	STRIKE	SCHLAGE S-SERIES	10-025	626		
	6	HINGE	McKINNEY	TA2714	US26D		
		SILENCER	GLYNN JOHNSON	GJ64			
	1	COORDINATOR			US26D		
D EXTERIOR DOOR	1	LOCKSET	SCHLAGE S-SERIES	JUPITER	626	PASSAGE	
	1	STRIKE	SCHLAGE S-SERIES	10-025	626		
	3	HINGE	McKINNEY	TA2714	US26D		
	1	CLOSER	NORTON	8301	US26D		
	1	WALLSTOP	IVES	WS447	US26D		
		SILENCER	GLYNN JOHNSON	GJ64			
	1	PERIMETER WEATHERSEAL	NATIONAL GUARD				

HARDWARE NOTES:
1. HARDWARE TYPES AND FUNCTIONS INDICATED IN THE HARDWARE SCHEDULE ARE THE BASIC INFORMATION NECESSARY FOR EACH DOOR OPENING. CONTRACTOR SHALL PROVIDE ALL HARDWARE NECESSARY FOR A COMPLETE INSTALLATION.
2. ALL HARDWARE SHALL MEET THE REQUIREMENTS OF ICC ANSI A117.1-2009.
3. GENERAL CONTRACTOR SHALL COORDINATE ALL CYLINDERS, LOCKSETS, AND KEYING WITH OWNER
4. COORDINATE FAIL SAFE LOCKS WITH REQUIREMENTS OF FIRE MARSHALL AND FIRE ALARM SYSTEM IN COMPLIANCE WITH IBC 1008.1.3

EQUIPMENT / PLUMBING FIXTURE SCHEDULE

Location	Description	Manufacturer	Model #	Finish	Style	Notes
MASTER BATH	TOILET	KOHLER	K3999-0	WHITE		
	SHOWER SET	PFISTER	PJX8310A			
		PFISTER	POX8340A			
	SINK	DURAVIT	D0305490000			
	SINK FAUCET	PFISTER	PGT49NC1C	CHROME		
	MIRROR	GATCO	SP-G4109S	CHROME		
GUEST BATH	TOILET	KOHLER	K3999-0	WHITE		
	SHOWER/TUB SET	MIRABELLE	MIRPR8030			
		MIRABELLE	MIRPR8020			
	SINK					
	SINK FAUCET	MIRABELLE	MIRWSPR800CP	CHROME	PROVINCETOWN	
	TUB	AMERICAN STANDARD	A2390202020	WHITE		
	MIRROR					FULL WALL CUSTOM CUT FLAT MIRRORS
ALL BATHS	VANITY CABINETS	LINEA STANDARD			BERTCH	WALL MOUNTED
	SHOWER DOORS					CUSTOM CUT HEAVY GLASS
	TOWEL BARS	KOHLER	K11371-CP			SIZE - 24"
	TOILET PAPER HOLDER	KOHLER	11374-CP			
	ROBE HOOKS	KOHLER	11375-CP			
KITCHEN	KITCHEN SINK	EMPIRE	[illegible]	STAINLESS		
	KITCHEN SINK FAUCET	KOHLER	[illegible]05-SN	POLISHED NICKEL		
	GARBAGE DISPOSAL	BADGER PRO	IPRO750			WITH AIR SWITCH IST500
	REFRIGERATOR	KITCHENAID	KRFC300ESS			COUNTER DEPTH
	DISHWASHER	KITCHENAID	KDTE104ESS			
	HOOD	KITCHENAID	KVWB400DSS			FOR UNITS 3-6
	MICROWAVE	KITCHENAID	KCMS1655BSS3			FOR UNITS 3-6
	GAS RANGE	KITCHENAID	KSGG700ESS			30"
	WASHER DRYER STACK	WHIRLPOOL	WFW70HEBW			
		WHIRLPOOL	WED70HEBW			
	MICROWAVE/HOOD	TBD	TBD			FOR UNITS 1-2
	KITCHEN CABINETS	WOLF HOME PRODUCTS		WHITE	DARTMOUTH	CLASSIC STANDARD CABINET, 1 1/4" FULL OVERLAY SHAKER STYLE
	KITCHEN KNOBS	TOP KNOBS	TM1121	CHROME	BERGEN	SIZE - 1 1/4"
	KITCHEN BAR PULLS	TOP KNOBS	TM1847	CHROME	HOPEWELL	
	KITCHEN BAR PULLS	TOP KNOBS	TM1849	CHROME	HOPEWELL	
	KITCHEN BAR PULLS	TOP KNOBS	TM1850	CHROME	HOPEWELL	

(TBS BY OWNER)

FINISH GROUP NOTES:
1. REFER TO DOOR SCHEDULE FOR DOOR AND DOOR FRAME FINISH
2. PROVIDE ALL INTERIOR FINISHES IN ACCORDANCE WITH THE FLAME SPREAD REQUIREMENTS IN 2012 IBC, ALL INTERIOR FINISHES TO COMPLY WITH IBC SMOKE DEVELOPMENT REQS.
3. ALL FINISHES AND EQUIPMENT SELECTIONS TO BE SELECTED BY OWNER, VERIFIED BY ARCHITECT
4. WOOD FLOORING INSTALLATION WILL COMPLY WITH IBC 803.11.1 OR 803.11.2

5105 LEE ST NE

ORGANIZATION FOR NEW ENVIRONMENTS LLC
dagi.abebe@gmail.com
Tel. 571.225.7211

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MATERIALS SCH.

A301

TABLE R402.4.1.1
AIR BARRIER AND INSULATION INSTALLATION

COMPONENT	CRITERIA[a]
Air barrier and thermal barrier	A continuous air barrier shall be installed in the building envelope. Exterior thermal envelope contains a continuous air barrier. Breaks or joints in the air barrier shall be sealed. Air-permeable insulation shall not be used as a sealing material.
Ceiling/attic	The air barrier in any dropped ceiling/soffit shall be aligned with the insulation and any gaps in the air barrier sealed. Access openings, drop down stair or knee wall doors to unconditioned attic spaces shall be sealed.
Walls	Corners and headers shall be insulated and the junction of the foundation and sill plate shall be sealed. The junction of the top plate and top of exterior walls shall be sealed. Exterior thermal envelope insulation for framed walls shall be installed in substantial contact and continuous alignment with the air barrier. Knee walls shall be sealed.
Windows, skylights and doors	The space between window/door jambs and framing and skylights and framing shall be sealed.
Rim joists	Rim joists shall be insulated and include the air barrier.
Floors (including above-garage and cantilevered floors)	Insulation shall be installed to maintain permanent contact with underside of subfloor decking. The air barrier shall be installed at any exposed edge of insulation.
Crawl space walls	Where provided in lieu of floor insulation, insulation shall be permanently attached to the crawlspace walls. Exposed earth in unvented crawl spaces shall be covered with a Class I vapor retarder with overlapping joints taped.
Shafts, penetrations	Duct shafts, utility penetrations, and flue shafts opening to exterior or unconditioned space shall be sealed.
Narrow cavities	Batts in narrow cavities shall be cut to fit, or narrow cavities shall be filled by insulation that on installation readily conforms to the available cavity space.
Garage separation	Air sealing shall be provided between the garage and conditioned spaces.
Recessed lighting	Recessed light fixtures installed in the building thermal envelope shall be air tight, IC rated, and sealed to the drywall.
Plumbing and wiring	Batt insulation shall be cut neatly to fit around wiring and plumbing in exterior walls, or insulation that on installation readily conforms to available space shall extend behind piping and wiring.
Shower/tub on exterior wall	Exterior walls adjacent to showers and tubs shall be insulated and the air barrier installed separating them from the showers and tubs.
Electrical/phone box on exterior walls	The air barrier shall be installed behind electrical or communication boxes or air sealed boxes shall be installed.
HVAC register boots	HVAC register boots that penetrate building thermal envelope shall be sealed to the sub-floor or drywall.
Fireplace	An air barrier shall be installed on fireplace walls. Fireplaces shall have gasketed doors.

- RECESSED LIGHT FIXTURE INSTALLED IN THE BUILDING THERMAL ENVELOPE SHALL BE AIR TIGHT, IC RATED, AND SEALED TO THE DRYWALL WITH DUCT SEALANT/ CAULK APPROVED FOR THE ELECTRICAL FIXTURES.

- MECHANICAL PIPING THAT IS CARRYING FLUIDS ABOVE 105 DEGREES OR BELOW 55 DEGREES WILL BE INSULATED TO AT LEAST R-3

- HVAC LINE SET PIPING WILL BE SHIELD FROM THE EFFECTS OF WEATHER BY USING POLYETHYLENE PIPE INSULATION

- FLOOR INSULATION R VALUE=R19
- WALL INSULATION R VALUE=R20
- CEILING INSULATION R VALUE=R49

- MECHANICAL VENTILATION OUTDOOR INTAKES AND EXHAUST VENTS WILL HAVE AUTOMATIC OR GRAVITY DAMPERS THAT CLOSE WHEN SYSTEM IS NOT IN OPERATION

- ATTIC ACCESS HATCH AND DOOR INSULATION SHALL=R VALUE OF R49

- A DUCT-LEAKAGE TEST MUST BE CONDUCTED. THE TOTAL DUCK LEAKAGE SHALL BE -8 CFM/100 SQ.FT. CONDITIONED FLOOR AREA AT PRESSURE OF 25 PASCAL, A WRITTEN REPORT OF THE RESULTS OF THE SHALL BE SIGNED BY THE PARTY CONDUCTIONG THE TEST AND PROVIDE TO THE OFFICIAL UPON REQUEST.

- MIN OF 85% OF LAMPS IN THE PERMANENTLY INSTALLED LIGHTING FIXTURES SHALL BE HIGH EFFICIENCY LAMPS OR MIN. OF 85% OF THE LIGHTING FIXTURES SHALL CONTAIN ONLY HIGH EFFICIENCY LAMPS.

EVAE BAFFLE ARE REQUIRED IN VENTED ATTIC ADJACENT TO SOFFIT OR EAVE. A MON. OF 1 INCH SHALL BE PROVIDED BETWEEN THE INSULATION AND ROOF SHEATHING AND AT THE LOCATION OF THE VENT. THE BAFFLE SHAL EXTEND OVER THE TOP OF THE INSULATION UNTIL IT IS AT LEAST 4 INCHES VERTICALLY ABOVE THE TOP OF THE INSULATION.
IN COMPLIANCE WITH IECC
402.2.3

NOTE:
AIR BARRIER & THE INSULATION/THERMAL BARRIER ARE ALLIGNED AND OTHER COMPONENTS OF THE BUILDINGS THERMAL ENVELOPE SHALL BE INSTALLED ACCORDING TO MANUFACTURER'S INSTRUCTIONS AND THE CRITERIA IN TABLE R402.4.1.1..

5105 LEE ST NE

ORGANIZATION FOR NEW ENVIRONMENTS LLC
dagi.abebe@gmail.com
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GENERAL INSULATION NOTES

A401

FLOOR-CEILING SYSTEMS, WOOD FRAMED

GA FILE NO. FC 5109 | **PROPRIETARY*** | **1 HOUR FIRE** | **55 to 59 STC SOUND**

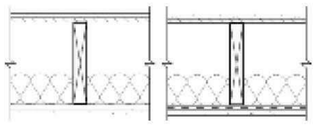
WOOD JOISTS, WOOD STRUCTURAL PANELS, GYPSUM FLOOR TOPPING, RESILIENT CHANNELS, GLASS OR MINERAL FIBER BATT OR LOOSE FILL INSULATION, GYPSUM WALLBOARD

One layer 5/8" proprietary type X gypsum wallboard or gypsum veneer base applied at right angles to resilient furring channels 24" o.c. (16" o.c. when batt insulation is used; 12" o.c. when loose fill insulation is used) with 1" Type S drywall screws 12" o.c. Gypsum board end joints located midway between continuous channels and attached with screws 8" to additional pieces of channel 60" long located 3" back on either side of end joint. Resilient channels applied at right angles to nominal 2 x 10 wood joists spaced a maximum of 24" o.c. with 1 1/4" Type S drywall screws. Glass or mineral fiber batt insulation stapled to subfloor or or loose fill insulation applied directly over gypsum board. Wood joists supporting 15/32" wood structural panel subfloor applied at right angles to joists with construction adhesive and 6d ring shank nails 12" o.c. Minimum 1/2" proprietary gypsum floor topping applied over subfloor.

STC and IIC rated with both joists and resilient channels spaced 16" o.c., 3 1/2" glass fiber insulation in joist spaces, 3/4" proprietary gypsum floor topping poured over 1/4" proprietary sound reduction mat, and with finish flooring of C&P, sheet vinyl, and engineered wood laminate.

PROPRIETARY GYPSUM COMPONENTS

United States Gypsum Company
- 5/8" SHEETROCK® Brand FIRECODE® C Core Gypsum Panels
- LEVELROCK® Brand Floor Underlayment



Approx. Ceiling Weight: 3 psf
Fire Test: UL R1319, 05NK04589, 2-4-05; UL R1319, 05NK09496, 3-31-05; UL Design L569
Sound Test: RAL TL04-97 & 98, 4-22-04; RAL TL04-99, - 100, -101, 4-26-04; RAL TL04-109, 4-30-04
IIC & Test: (73 generic C&P), RAL IN04-010, 4-22-04; (52 cushion sheet vinyl) RAL IN04-011, 4-22-04; (51 engineered wood laminate) RAL IN04-012, 4-26-04; (50 cushion sheet vinyl) RAL IN04-013, 4-26-04; (48 generic sheet vinyl) RAL IN04-014, 4-26-04; (45 cushion sheet vinyl & channels spaced 24" o.c.) RAL IN04-015, 4-30-04

GA FILE NO. WP 8111 | **PROPRIETARY*** | **1 HOUR FIRE**

GYPSUM WALLBOARD, GLASS MAT GYPSUM SUBSTRATE, RESILIENT CHANNELS, MINERAL OR GLASS FIBER INSULATION, WOOD STUDS

EXTERIOR SIDE: One layer 5/8" proprietary type X glass mat gypsum substrate (sheathing) applied parallel or at right angles to 2 x 4 wood studs 16" o.c. with 1 7/8" galvanized nails 7" o.c. 3" mineral or glass fiber insulation in stud space. Exterior cladding to be attached through glass mat gypsum substrate to studs.

INTERIOR SIDE: Resilient channels 24" o.c. attached at right angles to studs with one 1 1/4" Type W drywall screw at each stud. One layer 5/8" proprietary type X gypsum board applied at right angles to channels with Type S or S-12 drywall screws 8" o.c.

Joints staggered on opposite sides. **(LOAD-BEARING)**

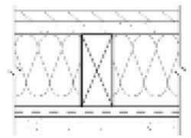
PROPRIETARY GYPSUM PANEL PRODUCTS

CertainTeed Gypsum Inc.
- 5/8" ProRoc® Type X Gypsum Panels
- 5/8" GlasRoc® Sheathing Type X Gypsum Panels

CertainTeed Gypsum Canada Inc.
- 5/8" ProRoc® Type X Gypsum Panels

National Gypsum Company
- 5/8" e²XP® FIRE-SHIELD® Gypsum Sheathing
- 5/8" Gold Bond® Brand FIRE-SHIELD® Gypsum Board

Temple-Inland
- 5/8" GreenGlass Type X
- 5/8" Type X

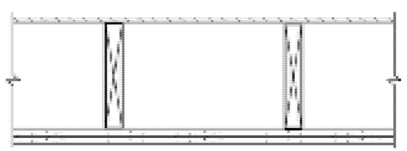


Thickness: 5 3/8"
Approx. Weight: 7 psf
Fire Test: UL R3660/R15187, 01NK21103, 2-4-02; UL R3501, 07NK17992, 12-12-07; UL R6937, 06NK17692, 9-19-08; UL Design U305

GA FILE NO. RC 2601 | **GENERIC** | **1 HOUR FIRE**

GYPSUM WALLBOARD, WOOD JOISTS, ROOF COVERING

Base layer 5/8" type X gypsum wallboard applied at right angles to 2 x 10 wood joists 24" o.c. with 1 1/4" Type W or S drywall screws 24" o.c. **Face** layer 5/8" type X gypsum wallboard or gypsum veneer base applied at right angles to joists with 1 7/8" Type W or S drywall screws 12" o.c. at joints and intermediate joists and 1 1/2" Type G drywall screws 12" o.c. placed 2" back on either side of end joints. Joints offset 24" from base layer joints. Wood joists supporting 1/2" plywood with exterior glue applied at right angles to joists with 8d nails. Appropriate roof covering. **Ceiling provides one hour fire resistance protection for framing, including trusses.**



Approx. Ceiling Weight: 5 psf
Fire Test: FM FC 172, 2-25-72; ITS, 8-6-98

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FIRE RELATED GENERAL DETAIL

A402

ELECTRICAL GENERAL NOTES

1. IT IS THE INTENT OF THESE DRAWINGS AND OTHER RELATED DOCUMENTS TO PRODUCE A COMPLETE AND FUNCTIONING ELECTRICAL SYSTEM. THE ELECTRICAL CONTRACTOR SHALL PROVIDE ALL LABOR, MATERIALS, TESTS, AND OTHER SERVICES AS MAY BE NECESSARY TO ACHIEVE THIS PRODUCT. THE CONTRACTOR SHALL ACKNOWLEDGE ACCEPTANCE OF THE PLANS AS AN ADEQUATE DEFINITION OF THE SCOPE OF WORK AND EXTRA COST CLAIMS BASED ON DISCREPANCIES ON THE PLANS WILL NOT BE CONSIDERED.
2. ALL ELECTRICAL WORK SHALL BE IN ACCORDANCE WITH THE NATIONAL ELECTRICAL CODE (NEC) AND ALL LOCAL CODES HAVING JURISDICTION. ALL EQUIPMENT, DEVICES, AND MATERIAL SHALL BE LISTED WITH UNDERWRITERS LABORATORIES FOR ITS APPLICATION AS INSTALLED AND SHALL BEAR THE UL LABEL.
3. THE ELECTRICAL CONTRACTOR SHALL OBTAIN ALL PERMITS AND PAY SUCH FEES AS MAY BE NECESSARY FOR INSPECTIONS, TESTS, AND OTHER SERVICES WHICH ARE REQUIRED FOR THE COMPLETION OF HIS WORK.
4. THE CONTRACTOR SHALL VISIT THE SITE AND EXAMINE CONDITIONS OF THE PREMISES AND THE CHARACTER AND EXTENT OF WORK REQUIRED PRIOR TO SUBMISSION OF BIDS. ANY DIFFICULTIES IN COMPLYING WITH THE DRAWINGS AND SPECIFICATIONS SHALL BE BROUGHT TO THE ATTENTION OF ARCHITECT BEFORE BIDDING.
5. ELECTRICAL PLANS ARE DIAGRAMMATIC. DO NOT SCALE DRAWINGS.
6. CONSULT PLANS OF ALL OTHER TRADES FOR COORDINATION AND FOR RELATED AND ADJOINING WORK.
7. CONSULT ARCHITECTURAL AND STRUCTURAL PLANS AND DETAILS FOR CONSTRUCTION TYPE, HEADROOM, ROOM FINISHES, CEILINGS, ETC.
8. SEE REFLECTED CEILING PLAN FOR EXACT LOCATION OF LIGHT FIXTURES.
9. CIRCUIT NUMBERS ARE FOR IDENTIFICATION PURPOSES ONLY. CONTRACTOR SHALL BE RESPONSIBLE FOR CORRECTLY SPACING THE CIRCUITS IN THE PANEL AND BALANCE THE LOAD ON THE PHASES UNDER NORMAL OPERATING CONDITIONS.
10. SHOP DRAWINGS FOR ALL ELECTRICAL EQUIPMENT, FIXTURES, DEVICES AND MATERIALS SHALL BE SUBMITTED TO THE ARCHITECT FOR APPROVAL BEFORE DELIVERY TO THE JOB SITE. EQUIPMENT, FIXTURES, DEVICES, AND MATERIAL DELIVERED TO THE JOB SITE OR INSTALLED PRIOR TO APPROVAL OF THE SHOP DRAWINGS, AND FOR WHICH THE SHOP DRAWINGS ARE SUBSEQUENTLY REJECTED, SHALL BE REPLACED WITH AN APPROVED ITEM AT NO ADDITIONAL COST TO THE OWNER.
11. CONTRACTOR SHALL VERIFY WIRE SIZES, C/B AND FUSE RATINGS FOR ALL HVAC EQUIPMENT, AND BRING TO THE ATTENTION OF THE ARCHITECT ANY DISCREPANCIES AFFECTING THE WORK PRIOR TO PROCEEDING.
12. ALL WORK SHALL BE DONE AT SUCH TIMES AND IN SUCH A MANNER AS WILL LEAST INTERFERE WITH THE MAINTENANCE AND OPERATION OF ALL RELATED OR AFFECTED SYSTEM. ALL POWER OUTAGES, FIRE ALARM SHUT DOWNS, ETC. SHALL BE COORDINATED WITH OWNER.
13. CONTRACTOR SHALL VERIFY THAT ALL DOOR SWINGS ARE CORRECT BEFORE INSTALLING LIGHT SWITCH OUTLETS.
14. CONTRACTOR SHALL BE RESPONSIBLE FOR PROPER SIZING OF ALL MOTOR OVERLOAD DEVICES (HEATERS) IN STARTERS BASED ON ACTUAL NAMEPLATE RATINGS ON THE MOTORS BEING INSTALLED.
15. HORSEPOWER RATINGS INDICATED ON DRAWINGS MAY DIFFER FROM ACTUAL EQUIPMENT FURNISHED. IF FURNISHED EQUIPMENT DIFFERS FROM RATINGS ON DRAWINGS, CONTRACTOR SHALL NOTIFY ARCHITECT/ENGINEER FOR APPROPRIATE ACTION TO BE TAKEN.
16. CONTRACTOR SHALL NOTE U.L. LABELS ON PACKAGED TYPE MECHANICAL EQUIPMENT. IF U.L. LABEL ON MECHANICAL EQUIPMENT OT ACTUALLY BE INSTALLED CALLS FOR THE OVERCURRENT PROTECTIVE DEVICE TO BE FUSES, THE ELECTRICAL CONTRACTOR SHALL PROVIDE A FUSED DISCONNECT SWITCH WITH PROPER SIZE FUSES AT THE SWITCH LOCATION INDICATED ON DRAWINGS AT NO ADDITIONAL CHARGE TO THE OWNER.
17. THE ELECTRICAL CONTRACTOR SHALL VERIFY THE TYPE OF CEILING SYSTEM WITH THE GENERAL CONTRACTOR OR CEILING CONTRACTOR TO INSURE THAT ALL RECESSED LIGHTING FIXTURES ARE COMPATIBLE WITH THE CEILING SYSTEM BEING INSTALLED. LIGHTING FIXTURES SHOULD NOT BE ORDERED UNTIL TYPE OF CEILING HAS BEEN VERIFIED.
18. LIGHTING FIXTURES INSTALLED IN SUSPENDED CEILINGS SHALL BE SUPPORTED DIRECTLY FROM THE BUILDING STRUCTURE.
19. THE CORRECT NUMBER OF WIRES MAY NOT BE INDICATED FOR ALL CIRCUITS, ONLY THOSE WHERE CLARIFICATION IS NECESSARY. THE ELECTRICAL CONTRACTOR SHALL PROVIDE ALL WIRES NECESSARY FOR THE PROPER FUNCTION OF THE SYSTEM WHETHER INDICATED ON DRAWINGS OF NOT.
20. ALL EMPTY CONDUIT RUNS IN EXCESS OF 10 FEET SHALL BE PROVIDED WITH A PULL WIRE OF FISH TAPE/CORD.
21. ALL CONDUCTORS, RACEWAYS AND CABLES SHALL BE CONCEALED IN CEILING OR WALL UNLESS INDICATED OTHERWISE.
22. OPENINGS OR CORE DRILLS IN EXISTING BUILDING STRUCTURE FOR PASSAGE OF CONDUITS/CABLES SHALL NOT BE CUT UNTIL THE CONTRACTOR HAS ASKED FOR AND RECEIVED WRITTEN APPROVAL FROM THE ARCHITECT AND OWNER.
23. THE LIGHTING FIXTURES SHALL BE FURNISHED AND INSTALLED COMPLETE WITH ALL ACCESSORIES (INCLUDING LAMPS) BY THE ELECTRICAL CONTRACTOR.
24. SYMBOLS SHOWN ON THIS SHEET ARE STANDARD SYMBOLS AND MAY NOT NECESSARILY ALL BE APPLICABLE TO THIS PROJECT.
25. THE CONTRACTOR SHALL GUARANTEE ALL HIS WORK AND MATERIALS FOR A PERIOD OF ONE YEAR AFTER ACCEPTANCE BY OWNER.
26. ALL PENETRATIONS OF FLOOR AND WALLS SHALL BE FIRE STOPPED IN ACCORDANCE WITH IBC, NEC, AND NFPA.
27. CONDUCTORS SHALL BE INSTALLED CONTINUOUS BETWEEN DEVICES, WITH SPLICES LOCATED ONLY IN JUNCTION BOXES OR IN CABINETS. CONDUCTORS SHALL BE OF SUFFICIENT LENGTH TO REACH THE FARTHEST TERMINAL IN PANELS. A MINIMUM OF 6" LOOPS SHALL REMAIN WHERE CONNECTIONS OF TAPS ARE TO BE MADE IN BRACH CIRCUIT WIRING.
28. PROVIDE AN UPDATED TYPEWRITTEN PANEL DIRECTORY IN EACH PANEL AFTER COMPLETION OF WORK.
29. ELECTRICAL CONTRACTOR SHALL PROVIDE AS BUILT DRAWINGS AND ALL MANUFACTURERS DATA AND WARRANTY LITERATURE AT THE COMPLETION OF THE CONTRACT.

WIRING DEVICES

1. THE ELECTRICAL CONTRACTOR SHALL COORDINATE ALL MOUNTING HEIGHTS FOR SWITCHES, RECEPTACLES, WALL MOUNTED LIGHT FIXTURES, AND TELEPHONE OUTLETS BY THE USE OF THE ARCHITECTURAL AND ENGINEERING DRAWINGS. SHOULD ANY CONFLICTS BECOME APPARENT THE CONTRACTOR SHALL REQUEST CLARIFICATION PRIOR TO INSTALLATION. IN THE WORK IS NOT COORDINATED ANY REMEDIAL WORK SHALL BE REDONE AT NO ADDITIONAL COST TO THE OWNER.
2. PROVIDE DUPLEX, SPECIFICATION GRADE RECEPTACLES 2 POLE, 3 WIRE GROUNDING WITH GREEN HEXAGONAL EQUIPMENT GROUND SCREWS, GROUND TERMINALS AND POLES INTERNALLY CONNECTED TO MOUNTING YOKE, 20 AMPERES, 125 VOLTS, WITH METAL PLASTER EARS, SIDE WIRING, NEMA CONFIGURATION 5–20R.
3. SWITCHES, 20 AMPS, 120/277 VOLTS, WITH MOUNTING YOKE INSULATED FROM MECHANISM, EQUIPPED WITH PLASTER EARS, SWITCH HANDLE, AND SIDE WIRED SCREW TERMINALS.
4. ALL SWITCHES, RECEPTACLES AND ASSOCIATED FACE PLATES SHALL BE OF WHITE COLOR. PREFERRED MANUFACTURER IS LUTRON.
5. ALL DEVICES INSTALLED IN THE LOCATION EXPOSED TO AMBIENT CONDITIONS SHALL BE WEATHERPROOFED.

EQUIPMENT SPECIFICATIONS

RACEWAY

1. MINIMUM SIZE OF THE CONDUIT SHALL BE 3/4". ALL RIDGE CONDUIT PER ARG.
2. PROVIDE FLEXIBLE CONDUIT FOR MOTOR CONNECTION, AND FOR OTHER ELECTRICAL EQUIPMENT CONDITION, WHERE SUBJECT TO MOVEMENT AND VIBRATION.
3. PROVIDE LIQUID TIGHT FLEXIBLE CONDUIT FOR CONNECTION OF MOTOR AND FOR OTHER ELECTRICAL EQUIPMENT WHERE SUBJECT TO MOVEMENT AND VIBRATION, AND ALSO WHERE SUBJECT TO ONE OR MORE OF THE FOLLOWING CONDITIONS, UNLESS NOTES OTHERWISE:
   A. MOIST AND HUMID ATMOSPHERE WHERE CONDENSATE CAN BE EXPECTED TO ACCUMULATE.
   B. CORROSIVE ATMOSPHERE
   C. SUBJECT TO DRIPPING OIL, GREASE OR WATER.
4. ALL CONDUITS SHALL BE GROUNDED PER NEC. CONDUITS ENTERING THE OUTLET BOXES, PANEL CABINETS, ETC. MUST BE FITTED WITH A DOUBLE LOCKNUT AND BUSHING.
5. PROVIDE RIGID STEEL, THREADED, THICK WALL CONDUIT, GALVANIZED OR EMT FOR ALL PANEL FEEDERS, AND ALL EXPOSED WIRING IN UNFINISHED AREAS. ALL RIDGE CONDUIT PER ARG.
6. ALL WIRE RACEWAYS IN OR PASSING THROUGH CONCRETE WALLS, SLABS, OR UNDERGROUND SHALL BE GALVANIZED RIGID STEEL THREADED CONDUIT.

WIRES AND CABLES

1. ALL WIRE AND CABLE SHALL BE COPPER WITH THHN/THWN INSULATION AND ALL WIRE SIZES ARE BASED ON COPPER CONDUCTORS WITH 75°C INSULATION UNLESS INDICATED OTHERWISE. ALL CONNECTORS, LUGS, ETC. SHALL BE LISTED FOR 75°C.
2. PROVIDE WIRING NOT SMALLER THAN #12 AWG FOR THE POWER DISTRIBUTION, AND NOT SMALLER THAN #14 AWG FOR THE FIRE ALARM SYSTEM.
3. ALL CIRCUITS 120/208 VOLT OVER 100 FEET AND ALL 277/480 VOLT CIRCUITS OVER 200 FEET FROM PANEL TO FIRST OUTLET SHALL HAVE CONDUCTORS ONE SIZE LARGER THAN NORMALLY REQUIRED WHETHER INDICATED ON PANEL SCHEDULE OR NOT.
4. CONDUCTORS INSTALLED UNDERGROUND OR IN THE WET LOCATIONS SHALL BE U.L. LISTED PER NEC, AND SHALL BE SUITABLE FOR WET LOCATIONS.

ELECTRICAL BOXES AND FITTINGS

1. ALL BOXES AND FITTINGS SHALL BE OF CODE–GAUGE STEEL.
2. JUNCTION AND PULL BOXES: PROVIDE GALVANIZED CODE–GUAGE STEET STEEL JUNCTION AND PULL BOXES WITH SCREW–ON COVER OF TYPES, SHAPES AND SIZES TO SUIT EACH RESPECTIVE LOCATION AND INSTALLATION, WITH WELDED SEAMS AND EQUIPPED WITH STAINLESS STEEL NUTS, SCREWS, AND WASHERS.
3. PROVIDE WEATHERPROOF OUTLETS FOR INTERIOR AND EXTERIOR LOCATIONS EXPOSED TO WEATHER OR MOISTURE.
4. ALL PULL BOXES SHALL BE FABRICATED FROM #12 OR HEAVIER GAUGE GALVANIZED STEEL AS REQUIRED BY THE NEC, AND SHALL BE EQUIPPED WITH SCREW FASTENED COVER.

PANELBOARDS

ALL PANELBOARDS MUST BE OF COPPER BUS ONLY.

PANELBOARD ENCLOSURES: PROVIDE GALVANIZED SHEET STEEL CABINET, CODE–GAUGE. CONSTRUCT WITH MULTIPLE KNOCKOUTS AND WIRING GUTTERS. PROVIDE FRONTS. EQUIPMENT WITH INTERIOR CIRCUIT DIRECTORY FRAME AND CARD WITH CLEAR PLASTIC COVERING. PROVIDE BAKED GRAY ENAMEL FINISH OVER A RUST INHIBITOR COATING. DESIGN ENCLOSURES FOR RECESSED MOUNTING. PROVIDE ENCLOSURES WITCH ARE FABRICATED BY SAME MANUFACTURES AS PANELBOARDS WHICH MATE AND MATCH PROPERLY WITH PANEL BOARD TO THE ENCLOSED. NEW PANELBOARDS SHALL BE AS INDICATED ON THE DRAWINGS, WITH BOLT–ON MOLDED CASE CIRCUIT BREAKERS AND COPPER BUS BARS. CIRCUIT BREAKERS AND INTERIORS SHALL BE OF THE SAME MANUFACTURER AND U.L. LISTED. THE PANELBOARD SHALL COMPLY WITH ALL APPLICABLE STANDARDS.

ALL BOXES MAY BE PLASTIC, RESIDENTIAL GRADE.

ALL WIRING DEVICES MAY BE RESIDENTIAL GRADE. CUTLER HAMMER OR APPROVED EQUAL.

MAY USE RESIDENTIAL MC CABLES.

FIRE RESISTANCE RATING NOTES

- ELECTRICAL BOXES PENETRATING FIRE RESISTANCE RATED MEMBRANES OF CEILINGS AND WALLS MUST MEET THE REQUIREMENTS OF 2013 DCBC CHAPTER 7 FOR PENETRATING ITEMS. THE BOXES THEMSELVES MUST BE FIRE RATED OR HAVE A FIRE STOP SYSTEM ADDED TO THEM (2013 DCBC 714.3.2)
- ALL ELECTRICAL WIRING, CONDUITS, RACEWAYS, CATV, AND TELEPHONE WIRING PENETRATING FIRE–RESISTANCE RATED MEMBRANES MUST BE PROPERLY SEALED TO ASSURE THAT ALL REQUIRED FIRE–RESISTANCE RATING IS NOT REDUCED PER 2013 DCBC 703.2. USE EVERKEM ECOSEAL FIRE STOP 814+ OR APPROVED EQUAL. IT IS UL CLASSIFIED, INTUMESCENT FIRE STOP SEALANT DESIGNED TO RESUME THE INTEGRITY OF 1 AND 2 HOUR RATED FIRE RATED WALLS AND FLOORS WHEN PENETRATED BY PLUMBING, ELECTRICAL, AND OTHER MECHANICAL ITEMS.

ELECTRICAL SYMBOLS

- LED LIGHTING FIXTURE. SEE LIGHTING FIXTURE SCHEDULE FOR TYPE AND MOUNTING.
- $ SINGLE POLE TOGGLE SWITCH – 20 AMP, 120 VOLT, M.H. 48" AFF, UNLESS NOTED OTHERWISE.
- $3 THREE WAY TOGGLE SWITCH – 20 AMP, 120 VOLT, M.H. 48" AFF, UNLESS NOTED OTHERWISE.
- $M MOTOR RATED SWITCH WITH THERMAL OVERLOAD PROTECTION, MOUNT ADJACENT TO OR ON THE MOTOR BEING CONTROLLED.
- SC SMOKE/CARBON MONOXIDE DETECTOR
- $D DIMMER SWITCH
- J JUNCTION BOX, CEILING OR WALL MOUNTED.
- DUPLEX GROUNDING TYPE RECEPTALCE – 20 AMP, 120 VOLT, NEMA 5–20R, M.H. 18" AFF, UNLESS NOTED OTHERWISE.
- DOUBLE DUPLEX GROUNDING TYPE RECEPTACLE – 20 AMP, 120 VOLT, NEMA 5–20R, M.H. 18" AFF, UNLESS NOTED OTHERWISE.
- DUPLEX RECEPTACLE WITH GROUND FAULT CIRCUIT INTERRUPTER PROTECTION – 20 AMP, 120 VOLT, NEMA 5–20R, M.H. 42" AFF, UNLESS NOTED OTHERWISE.
- PANELBOARD
- DISCONNECT SWITCH, AMP, VOLT, POLES AND FUSING AS NOTED ON DRAWING.
- ELECTRIC MOTOR CONNECTION – HORSEPOWER AS NOTED.
- M POWER COMPANY METER, RATING AS INDICATED ON DRAWING.
- 2 HOMERUN WIRING WITH A DEDICATED EQUIPMENT GROUND WIRE TO THE REFERENCED PANELBOARD. ARROW HEADS AND NUMERALS INDICATE THE CIRCUIT NUMBERS.

ELECTRICAL ABBREVIATIONS

A	AMP	KW	KILOWATTS
AFF	ABOVE FINISHED FLOOR	MCB	MAIN CIRCUIT BREAKER
AHU	AIR HANDLING UNIT	MLO	MAIN LUGS ONLY
AIC	AMPS INTERRUPTING CAPACITY	MOCP	MAXIMUM OVERCURRENT PROTECTION
AL	ALUMINUM	N	NEW
ATS	AUTOMATIC TRANSFER SWITCH	NEC	NATIONAL ELECTRICAL CODE
AFG	ABOVE FINISHED GRADE	NEMA	NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION
AWG	AMERICAN WIRE GAUGE		
CB	CIRCUIT BREAKER	NFPA	NATIONAL FIRE PROTECTION ASSOCIATION
CH	CABINET HEATER	NFSS	NON–FUSED SAFETY SWITCH
CKT	CIRCUIT	OCPD	OVER CURRENT PROTECTION DEVICE
CU	COPPER, COMPRESSOR UNIT	P	POLE
C	CONDUIT	PH OR ø	PHASE
DN	DOWN	PNL	PANEL
E	EXISTING	PFF	PROVISIONS FOR FUTURE
EC	EMPTY CONDUIT	R	REMOVE
ER	EXISTING RELOCATED	RE	RELOCATE
EF	EXHAUST FAN	RTU	ROOF TOP UNIT
EM	EMERGENCY	RLA	RUNNING LOAD AMPS
EMT	ELECTRIC METAL TUBING	SN	SOOLID NEUTRAL
EPO	EMERGENCY POWER OFF	SWED	SWITCHBOARD
EWC	ELECTRIC WATER COOLER	TELE	TELEPHONE
FSS	FUSED SAFETY SWITCH	TYP	TYPICAL
FACP	FIRE ALARM CONTROL PANEL	UL	UNDERWRITERS LABORATORY
FAAP	FIRE ALARM ANNUNCIATOR PANEL	UON	UNLESS OTHERWISE NOTED
FLA	FULL LOAD AMPS	V	VOLTAGE
GFI	GROUND FAULT INTERRUPTER	VA	VOLT AMPS
GND OR G	GROUND	VAV	VARIABLE AIR VOLUME
HP	HORSE POWER	W	WATTS
HG	HOSPITAL GRADE	WH	WATER HEATER
IG	ISOLATED GROUND	WP	WEATHER PROOF
JB	JUNCTION BOX	WSA	WIRE SIZE AMPS
KVA	KILOVOLT AMPS	XFMR	TRANSFORMER

WASHINGTON DC CODES

BUILDING:	2015 IBC AND 2017 DCMAR12A SUPPLEMENT 2015 IEBC
COVERING CODES:	2017 DISTRICT OF COLUMBIA BUILDING CODE
MECHANICAL:	2017 DISTRICT OF COLUMBIA MECHANICAL CODE
ELECTRICAL:	2014 INTERNATIONAL NFPA–70
PLUMBING:	2017 DISTRICT OF COLUMBIA PLUMBING CODE
FIRE:	2017 DISTRICT OF COLUMBIA FIRE CODE
GREEN:	2017 DISTRICT OF COLUMBIA GREEN CONSTRUCTION CODE
ENERGY:	2017 DISTRICT OF COLUMBIA ENERGY CONSTRUCTION CODE

SHEET INDEX

DRAWING NUMBER	DRAWING TITLE
E100	ELECTRICAL COVER SHEET
E200	ELECTRICAL LIGHTING FLOOR PLANS
E300	ELECTRICAL POWER FLOOR PLANS
E400	ELECTRICAL PANEL SCHEDULES AND RISER DIAGRAM

5105 LEE STREET NE
DISTRICT OF COLUMBIA
20019

ONE DESIGN SERVICES
organizationfornewenvironment@gmail.com
Tel. 571.225.7211

DISTRICT OF COLUMBIA · KHALID JAMAAFI · PE022123 · PROFESSIONAL ENGINEER
12/29/2022

ENGINEER: KHALID LAMAAFI
ADDRESS: 9707 ASPEN PL.
MANASSAS VA, 20110
EMAIL: ALLATMEP@GMAIL.COM
CELL: 571-833-8357

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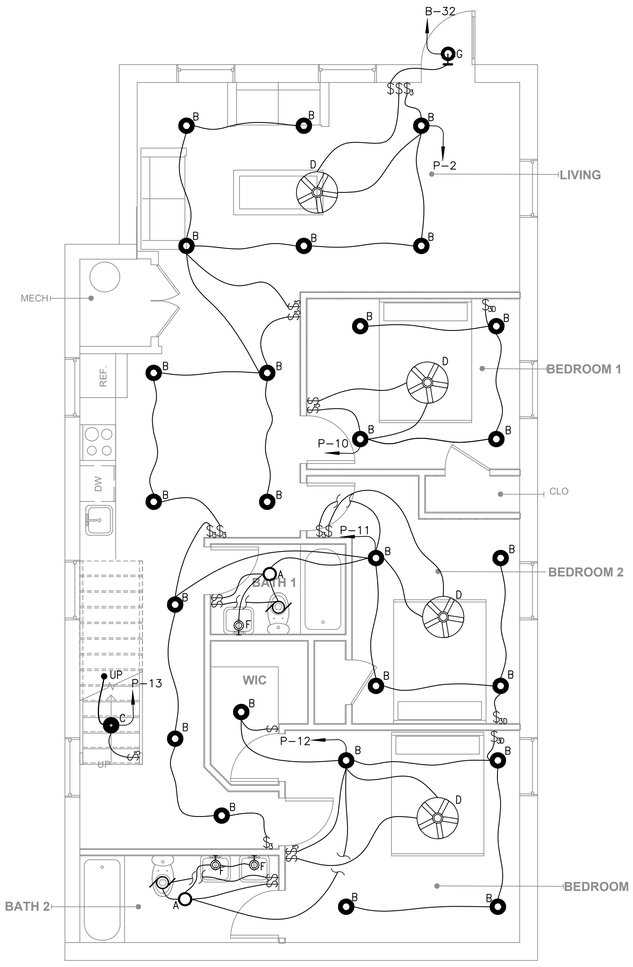
ELECTRICAL COVER SHEET

E100

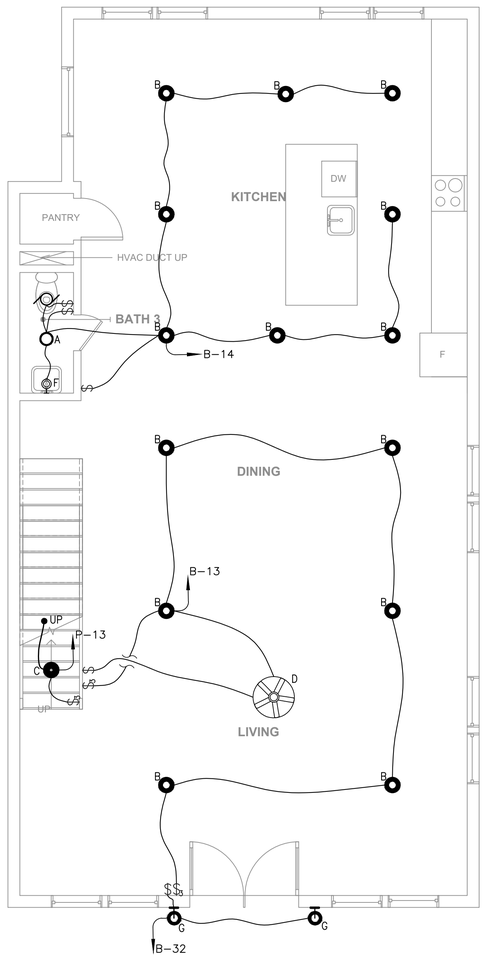
LIGHTING FIXTURE SCHEDULE

AFF ABOVE FINISHED FLOOR
BFC BELOW FINISHED CEILING
C CEILING
DL DOWN LIGHT
EB EMERGENCY BATTERY
F FLUORESCENT
PL POLE
P PENDANT
R RECESSED
S SURFACE
U UNIVERSAL
W WALL

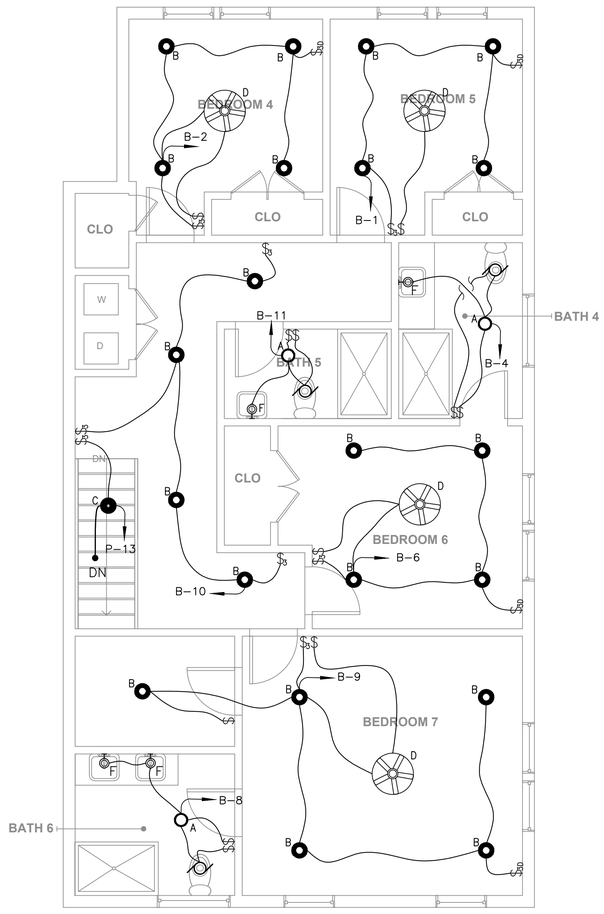
	FIXTURE TYPE	MANUFACTURER	CATALOG No.	LAMP No.	LAMP TYPE	VOLTS	MOUNTING	LOCATION	REMARKS	LIMENS	LUMENS/W	HIGH EFFICIENCY(Y,N)	QUANTITY	TOTAL WATTS
A		PROGRESS	P810016-030-30	1	17W LED	120	S	TOILETS	ROUND LED	1325	52	Y	6	102
B		PROGRESS	P810029-020-30	1	16W LED	120	S	AS SHOWN	LED ADJUSTABLE FLUSH LIGHT	950	61	Y	61	976
C		BESA LIGHTING	PELLA1307C	1	16W LED	120	S	STAIRCASE	LED SURFACE ROUND DIMMABLE	1500	94	Y	3	48
D		PROGRESS	250036-129-30	1	24W LED	120	P	AS SHOWN	LED LIGHT AND CEILING FAN	1500	94	Y	9	216
G		PROGRESS	P5676-20 30K9	1	9W LED	120	S	EXTERIOR	LED SECURITY LIGHT	623	69	Y	3	27
F		PROGRESS	P2145-1530K9	1	9W LED	120	S	TOILETS	VANITY LED LIGHT	623	69	Y	8	72
										TOTAL FIXTURES: 77 EFFICIENCY FIXTURES: 100%				TOTAL: 1441 W 1441W / 3530 SF = 0.40



1 ELECTRICAL LIGHTING CELLAR FLOOR PLAN
SCALE: 1/4"=1'-0"



2 ELECTRICAL LIGHTING FIRST FLOOR PLAN
SCALE: 1/4"=1'-0"



3 ELECTRICAL LIGHTING SECOND FLOOR PLAN
SCALE: 1/4"=1'-0"

GENERAL APARTMENT NOTES:

1. ALL SMOKE DETECTORS IN UNITS ARE TO BE 110 VOLT WITH AUDIBLE ALARM AND BATTERY BACKUP AND INTERCONNECTED AND SIMULTANEOUSLY ALARM WHEN ANY SINGLE DETECTOR IS ACTIVATED.
2. ALL 20A/1 AND 15A/1 POLE CIRCUIT BREAKERS EXCEPT EXTERIOR SHALL BE AFCI TYPE. PER 2017 DCEC 2017.12.
3. ALL KITCHEN (15A & 20A, 120 VOLT), TOILET AND EXTERIOR OUTLETS AND WITHIN 6' OF WATER SOURCE SHALL BE GFCI TYPE OR CIRCUIT BREAKER TO BE GFCI TYPE.
4. ALL KITCHEN,TOILET,AND EXTERIOR RECEPTACLES SHALL BE GFCI TYPE.
5. COORDINATE IN THE FIELD FOR EXACT LOCATION OF ALL MECHANICAL UNITS.
6. INSTALL TAMPER RESISTANCE NON-LOCKING TYPE RECEPTACLES FOR ALL NON-FIXED APPLIANCE RECEPTACLES.
7. DISHWASHER CIRCUIT SHALL BE SUPPLIED BY GFCI RECEPTACLE. ACCESSIBLY LOCATED UNDER THE SINK.
8. ALL WIRING DEVICES TO BE RESIDENTIAL GRADE. CUTLER HAMMER OR APPROVED EQUAL.
9. MAY USE APPROVED RESIDENTIAL MC CABLE OR ROMEX IN LIEU OF CONDUITS FOR APARTMENTS.
10. COMPLY WITH AUTHORITIES HAVING JURISDICTION FOR RESIDENTIAL WIRING.
11. LOCATE DISPOSAL JUNCTION BOX 24" AFF UNDER SINK CONTROLLED BY MOTOR RATED SWITCH.
12. DUPLEX RECEPTACLES CIRCUITS: MOUNT 6" ABOVE COUNTERS IN KITCHENS AND TOILETS AND 18" AFF IN OTHER AREAS.
13. COMPLY WITH ADA REQUIREMENTS FOR DEVICE HEIGHTS.
14. DIMMER SWITCHES THROUGHOUT UNIT EXCEPT TOILETS.
15. A MINIMUM 85% OF LAMPS IN PERMANENTLY INSTALLED LIGHTING FIXTURES SHALL BE HIGH EFFICACY LAMPS PER SECTION N1104.

FIRE RESISTANCE RATING NOTES:

1. ELECTRICAL BOXES PENETRATING FIRE RESISTANCE RATED MEMBRANES OF CEILINGS AND WALLS MUST MEET THE REQUIREMENTS OF 2017 DCBC CHAPTER 7 FOR PENETRATING ITEMS. THE BOXES THEMSELVES MUST BE FIRE RATED OR HAVE A FIRE STOP SYSTEM ADDED TO THEM (2017 DCBC 714.3.2)
2. ALL ELECTRICAL WIRING, CONDUITS, RACEWAYS, CATV, AND TELEPHONE WIRING PENETRATING FIRE-RESISTANCE RATED MEMBRANES MUST BE PROPERLY SEALED TO ASSURE THAT ALL REQUIRED FIRE-RESISTANCE RATING IS NOT REDUCED PER 2017 DCBC 703.2. USE EVERKEM ECOSEAL FIRE STOP 814+ OR APPROVED EQUAL. IT IS UL CLASSIFIED, INTUMESCENT FIRE STOP SEALANT DESIGNED TO RESUME THE INTEGRITY OF 1 AND 2 HOUR RATED FIRE RATED WALLS AND FLOORS WHEN PENETRATED BY PLUMBING, ELECTRICAL, AND OTHER MECHANICAL ITEMS.

5105 LEE STREET NE
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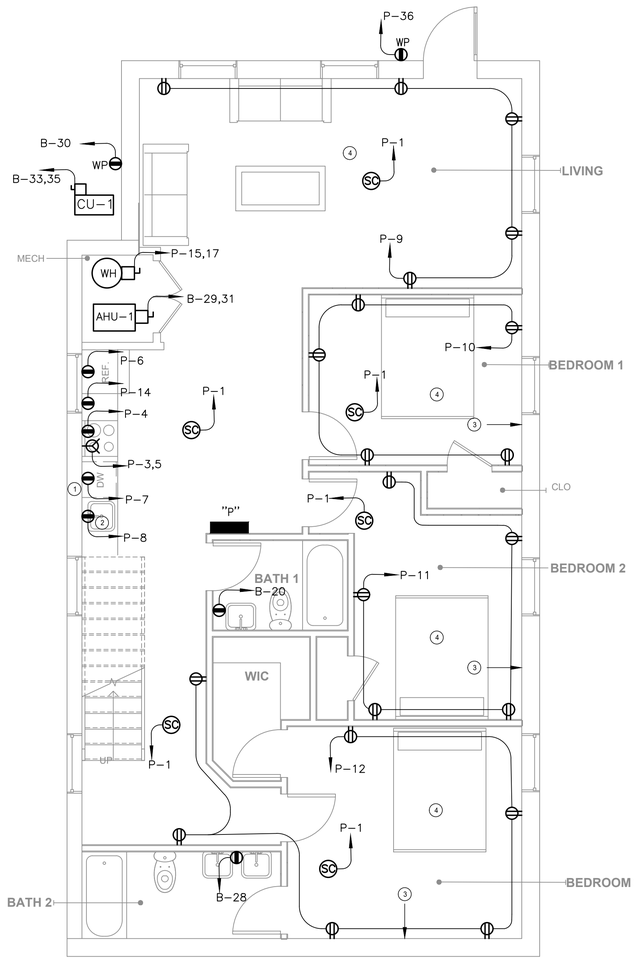
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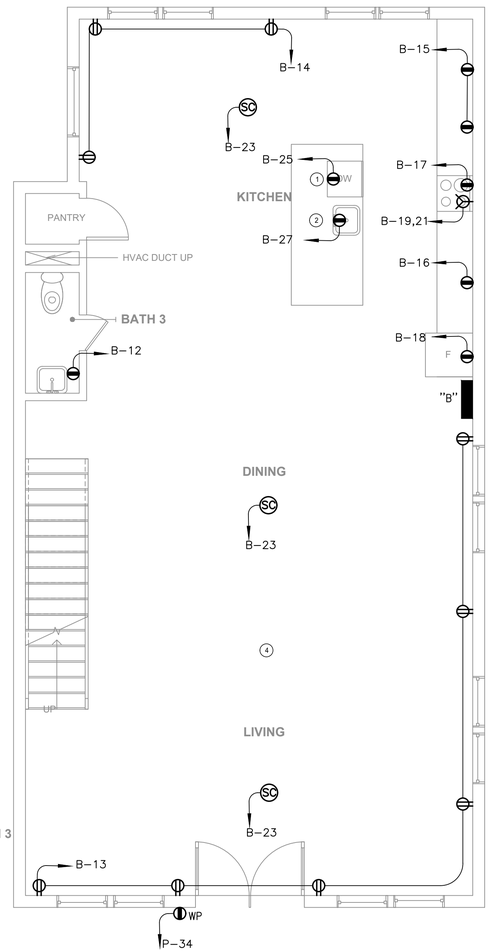
ELECTRICAL LIGHTING FLOOR PLANS

E200

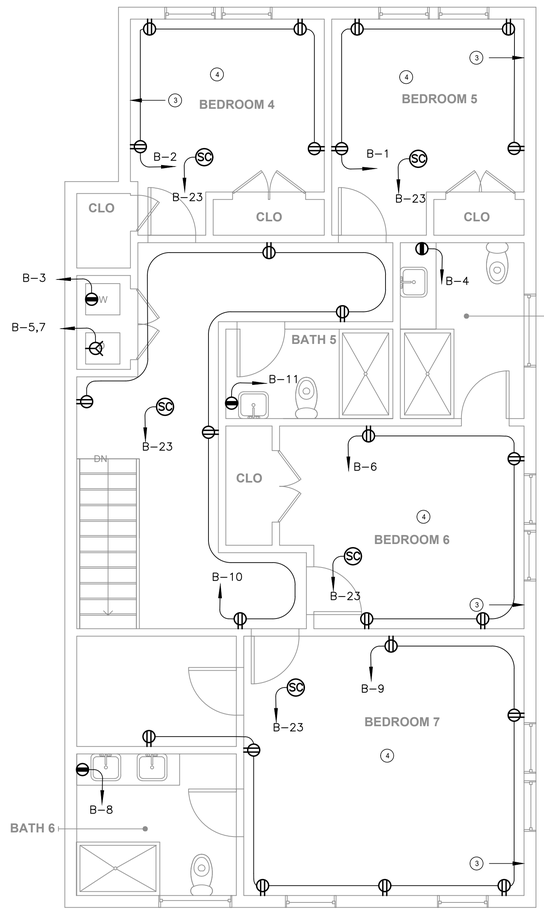
MECHANICAL EQUIPMENT SCHEDULE								
LOAD	V	PH	FLA	MCA	MOCP	FUSE	DISC.SWITCH	NEMA
AHU-1	240	1	34,4	43	70	70	60	1
CU-1	240	1	17,2	21,5	30	30	30	3R



1 ELECTRICAL POWER CELLAR FLOOR PLAN
SCALE: 1/4"=1'-0"



2 ELECTRICAL POWER FIRST FLOOR PLAN
SCALE: 1/4"=1'-0"



3 ELECTRICAL POWER SECOND FLOOR PLAN
SCALE: 1/4"=1'-0"

POWER KEYED NOTES

1. DISHWASHER RECEPTACLE MUST BE ACCESSIBLE UNDER THE SINK.
2. DISPOSAL RECEPTACLE MUST BE ACCESSIBLE UNDER THE SINK.
3. FIRE RATED WALL THROUGHOUT THE PERIMETER WALL.
4. FIRE RATED CEILING THROUGHOUT THE UNIT.

GENERAL APARTMENT NOTES:

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13. COMPLY WITH ADA REQUIREMENTS FOR DEVICE HEIGHTS.
14. DIMMER SWITCHES THROUGHOUT UNIT EXCEPT TOILETS.
15. A MINIMUM 85% OF LAMPS IN PERMANENTLY INSTALLED LIGHTING FIXTURES SHALL BE HIGH EFFICACY LAMPS PER SECTION N1104.
16. RANGE, 50A, 240V, 3#6 + 1#10 GROUND IN 3/4" CONDUIT, CONNECT TO EP*/21,23, PROVIDE A RANGE RECEPTACLE WITH MATCHING CORD & PLUG MOUNTED 24" A.F.F.
17. RECEPTACLE FOR WASHER/DRYER , MH 2'-0" AFF, 3#10 + 1#10 GROUND IN 3/4" CONDUIT CONNECT DRYER TO EP*/17,19, AND WASHER TO CIRCUIT EP*/15.

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ONE DESIGN SERVICES
organizationfornewenvironment@gmail.com
Tel. 571.225.7211

DISTRICT OF COLUMBIA · KHALID LAMAAFI · PE922123 · PROFESSIONAL ENGINEER
12/29/2022

ENGINEER: KHALID LAMAAFI
ADDRESS: 9707 ASPEN PL.
MANASSAS VA, 20110
EMAIL: ALLATMEP@GMAIL.COM
CELL: 571-833-8357

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ELECTRICAL POWER FLOOR PLANS

E300

NEW PANELBOARD SCHEDULE — B

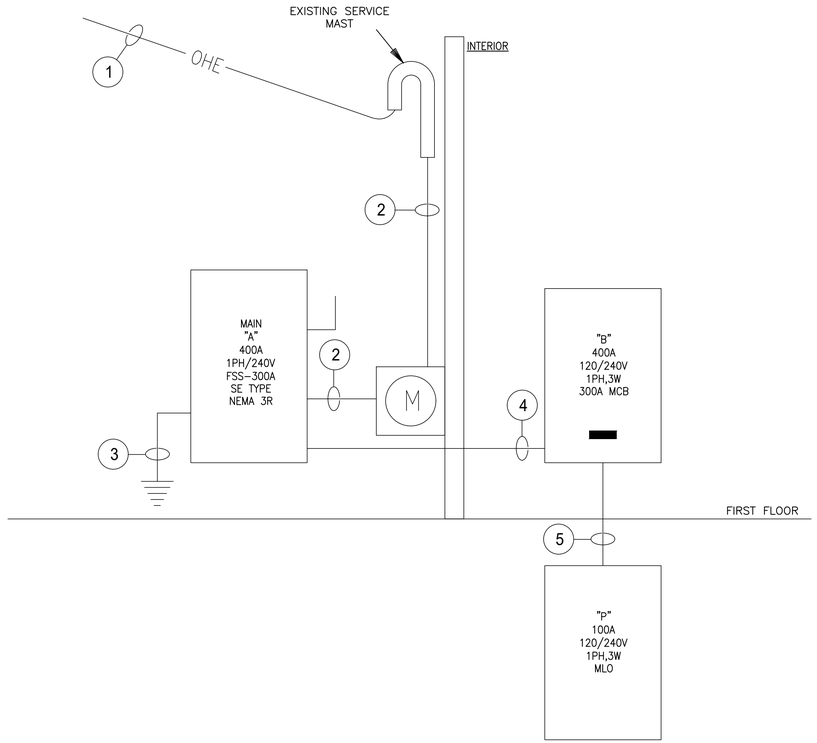
LOCATION: FIRST FLOOR

C.B. RATING: 22 K.A.I.C.

WIRE SIZE	TYPE	VOLTAGE	PHASE	WIRE		MOUNTING		SIZE	LUG	TYPE	TYPE	WIRE SIZE
		240/120V	1	3		SURFACE		400	MCB	NEMA 1		
		USE and/or AREA SERVED		C/B POLE	CIR	LOAD ØA	LOAD ØB	CIR	C/B POLE	USE and/or AREA SERVED		
		RECEPTACLES, LIGHTING BEDROOM 5		20/1	1	700						
						800		2	20/1	RECEPTACLES, LIGHTING BEDROOM 4		
		WASHER		20/1	3		1200					
							1000	4	20/1	RECEPTACLES, LIGHTING BATH 4		
		DRYER		30/2	5	1500						
						800		6	20/1	RECEPTACLES, LIGHTING BEDROOM 6		
					7		1500					
							1000	8	20/1	RECEPTACLES, LIGHTING BATH 6		
		RECEPTACLES, LIGHTING BEDROOM 7		20/1	9	1300						
						1000		10	20/1	RECEPTACLES, LIGHTING HWY		
		RECEPTACLES, LIGHTING BATH 5		20/1	11		1000					
							800	12	20/1	RECEPTACLES GFCI BATH 3		
		RECEPTACLES, LIGHTING LIVING AND DINING		20/1	13	1100						
						700		14	20/1	RECEPTACLES, LIGHTING KITCHEN AND LGT BATH 3		
		APPLIANCE		20/1	15		800					
							800	16	20/1	APPLIANCE		
		MICROWAVE		20/1	17	1200						
						1200		18	20/1	REFRIGERATOR		
		STOVE		50/2	19		4000					
							800	20	20/1	RECEPTACLES GFCI BATH 1		
					21	4000						
						7000		22	100/3	SUB PANEL P		
		SMOKE/CARBONG MONOXIDE DETECTORS		20/1	23		400					
							7000	24				
		DISHWASHER		20/1	25	1000						
						7000		26				
		DIPOSAL		20/1	27		600					
							800	28	20/1	RECEPTACLES GFCI BATH 2		
		AHU-1		70/2	29	4200						
						1000		30	20/1	RECEPTACLES GFCI EXTERIOR		
					31		4200					
							100	32	20/1	LIGHTING EXTERIOR		
		CU-1		30/2	33	2100						
						1200		34	20/1	RECEPTACLE EXTERIOR		
					35		2100					
							1200	36	20/1	RECEPTACLE EXTERIOR		
		TOTAL LOAD PER PHASE				37800	29300			TOTAL LOAD PER PHASE		

FEEDER OCPD AND CONDUCTOR CALCULATION

LOAD DESCRIPTION (LOAD IN KVA)	CONNECT LOAD	DEMAND LOAD	NOTES
TOTAL KVA	67	37,00	
TOTAL AMPS	280 A	192 A	



POWER RISER DIAGRAM

N.T.S

NEW PANELBOARD SCHEDULE — P

LOCATION: CELLAR FLOOR

C.B. RATING: 22 K.A.I.C.

WIRE SIZE	TYPE	VOLTAGE	PHASE	WIRE		MOUNTING		SIZE	LUG	TYPE	TYPE	WIRE SIZE
		240/120V	1	3		SURFACE		100	MLO	NEMA 1		
		USE and/or AREA SERVED		C/B POLE	CIR	LOAD ØA	LOAD ØB	CIR	C/B POLE	USE and/or AREA SERVED		
		SMOKE/CARBONG MONOXIDE DETECTORS		20/1	1	200						
						170		2	20/1	LIGHTING LIVING AND KITCHEN		
		STOVE		50/2	3		4000					
							1100	4	20/1	MICROWAVE		
					5	4000						
						1100		6	20/1	REFRIGERATOR		
		DISHWASHER		20/1	7		1000					
							600	8	20/1	DIPOSALE		
		RECEPTACLES LIVING		20/1	9	700						
						700		10	20/1	RECEPTACLES AND LIGHTING BEDROOM 1		
		RECEPTACLES, LIGHTING BEDROOM 2, LGT BATH 1 AND HWY		20/1	11		1000					
							1200	12	20/1	RECEPTACLES, LIGHTING BEDROOM 3, RECEPTACLES HWY AND LGT BATH 2		
		LIGHTING STAIRS		20/1	13	200						
						1000		14	20/1	APPLIANCE		
		WATER HEATER		30/2	15		2500					
							-	16	-	SPACE		
					17	2500						
						-		18	-	SPACE		
		TOTAL LOAD PER PHASE				10570	11400			TOTAL LOAD PER PHASE		

FEEDER OCPD AND CONDUCTOR CALCULATION

LOAD DESCRIPTION (LOAD IN KVA)	CONNECT LOAD	DEMAND LOAD	NOTES
TOTAL KVA	22	27,50	
TOTAL AMPS	92 A	114A	

LOAD CALCULATION

		UNIT 1
1. A. GENERAL LOAD	.....SF x 3 VA	3530 S.F x 3 VA = 10590 VA
B. SMALL APPLIANCE LOADS	1500 VA x 2 CIRCUITS	1500 x 2 VA = 3000 VA
C. LAUNDRY LOAD	1200 VA x 1	1200 x 1 VA = 1200 VA
	SUBTOTALE	14790 VA
FIRST	3000 VA x 100%	3000 VA
NEXT	..... VA x 40%	11790 VA x 40 % = 4716 VA
	TOTAL	= 7716
2.COOKING	8000 VA x 1	8000 VA
3.DRYER	3000 VA x 1	3000 VA
4. FIXED APPLIANCE LOADS		
DISPOSAL	800 VA x 1	800 VA
APPLAINCES	3000 VA x 1	3000 VA
WATER HEATER	6000 VA x 1	6000 VA
5. HEATING OR COOLING LOAD		
HEATING	8000 VA x 1	8000 VA
COOLING	3000 VA x 1	3000 VA
LARGEST LOAD		8000 VA
6. LARGES MOTOR LOAD	1600 VA x 25%	400 VA
	TOTAL	36,916 VA
		36,916 / 240 V = 154 AMP
		154 A x 1.25 = 192 AMP
		NEW SERVICE 400A @240/1PH

POWER RISER NOTES

1. 400 AMP OVERHEAD ELECTRICAL SERVICE.
2. 3#350 MCM, 3" CONDUIT.
3. GROUND PER NEC 250. NO.2 GND., TO GROUND ROD, STREET SIDE OF WATER MAIN AND FOUNDATION REBAR.
4. 3#350 MCM, 1#2 GND., 3" CONDUIT.
5. 3#3, 1#6 GND., 1-1/2" CONDUIT.

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ELECTRICAL PANEL SCHEDULES AND RISER DIAGRAM

E400

MECHANICAL SPECIFICATION

DIVISION 23 – HEATING, VENTILATING AND AIR CONDITIONING

23A.1 GENERAL

A. THE GENERAL CONDITIONS, SUPPLEMENTARY CONDITIONS AND APPLICABLE PROVISIONS OF OTHER DIVISIONS, FORM A PART OF THIS SPECIFICATION AND CONTRACT, AND SHALL BE CAREFULLY EXAMINED BY EACH BIDDER BEFORE SUBMITTING HIS PROPOSAL.

B. COMPLIANCE WITH LOCAL JURISDICTIONS: ALL WORK PERFORMANCE UNDER THIS SECTION SHALL CONFORM TO THE REQUIREMENTS OF DRAWINGS AND SPECIFICATIONS AND TO THE CODES, ORDINANCES AND STANDARDS OF THE LOCAL JURISDICTION. IN CASE OF A CONFLICT BETWEEN DRAWINGS OR SPECIFICATIONS AND THE REQUIREMENT OF THE LOCAL JURISDICTION, THE MORE STRINGENT REQUIREMENTS SHALL APPLY.

C. THIS CONTRACTOR SHALL OBTAIN AND PAY FOR ALL PERMITS AND FEES FOR INSPECTIONS RELATED TO HIS WORK.

D. INSTALLATION SHALL BE GUARANTEED FOR WORKMANSHIP, MATERIALS AND EQUIPMENT AGAINST DEFECTS, LEAKS, SYSTEM PERFORMANCE OR NON–OPERATION FOR A PERIOD OF ONE YEAR AFTER DATE OF ACCEPTANCE. CONTRACTOR SHALL PAY ALL COSTS INVOLVING THE GUARANTEE OF ALL SYSTEMS.

E. BEFORE ANY MATERIALS OR EQUIPMENT IS PURCHASED, THIS CONTRACTOR SHALL SUBMIT TO THE OWNER FOR APPROVAL 6 SETS OF SUBMITTAL DATA. THIS DATA SHALL INCLUDE CATALOG NUMBERS, CUTS, DIAGRAMS, DRAWINGS AND OTHER DESCRIPTIVE DATA AS MAY BE REQUIRED FOR ALL EQUIPMENT TO BE INSTALLED. INCLUDE SHEET METAL FABRICATION, SHOP DRAWINGS AND LAYOUTS COORDINATED WITH EXISTING CONDITIONS.

F. THE WORD "PROVIDE" AS USED IN SPECIFICATIONS AND ON PLANS, SHALL MEAN FURNISH AND INSTALL COMPLETE AND READY FOR USE.

G. ALL WORK SHALL BE COORDINATED WITH ALL TRADES PRIOR TO INSTALLATION.

H. DO NOT CUT STRUCTURAL MEMBERS WITHOUT THE APPROVAL OF THE ARCHITECT, AND ALL SUCH CUTTING SHALL BE DONE IN A MANNER DIRECTED BY HIM.

I. IN GENERAL, DRAWINGS FOR THE WORK ARE DIAGRAMMATIC AND SHOW THE LOCATION, TYPE AND SIZE OF PIPING, EQUIPMENT, DUCTWORK AND ACCESSORY EQUIPMENT. THE CONTRACTOR SHALL FURNISH ALL ITEMS NECESSARY FOR THE PROPER INSTALLATION OF THE WORK. THE CONTRACTOR SHALL VERIFY ALL NECESSARY DIMENSIONS BEFORE INSTALLING ANY OF THE WORK, AND SHALL CHECK HIS LAYOUTS TO ALLOW CLEARANCE REQUIRED FOR OTHER WORK AS SHOWN ON THE RAWINGS. THE SCOPE OF WORK CONSISTS GENERALLY OF PROVIDING AND INSTALLING THE DUCTWORK DISTRIBUTION SYSTEM, AND FINAL PREPARATION FOR TESTING OF ALL SYSTEMS AND EQUIPMENT.

23A.2 PRODUCTS AND INSTALLATION

A. NOT USED.

1. MANUAL DAMPERS: PROVIDE MANUAL BALANCING DAMPERS AS MANUFACTURED BY RUSKIN OR APPROVED EQUAL WITH LOCKING QUADRANT ON DUCTS AT LOCATIONS SHOWN.

2. SPLITTER DAMPERS: AT ALL DUCT BRANCHES, PROVIDE SPLITTER DAMPERS WITH ROD AND SET SCREW IN ACCORDANCE WITH SMACNA STANDARDS.

3. ALL LONGITUDINAL AND TRANSVERSE JOINTS, SEAMS AND CONNECTIONS OF SUPPLY AND RETURN DUCTS OPERATING AT A STATIC PRESSURE LESS THAN OR EQUAL TO 2 INCHES WATER GAUGE (W.G.) (500 PA) SHALL BE SECURELY FASTENED AND SEALED WITH WELDS, GASKETS, MASTICS (ADHESIVES), OR MASTIC–PLUS–EMBEDDED–FABRIC SYSTEMS IN ACCORDANCE WITH THE MANUFACTURER'S INSTALLATION INSTRUCTIONS. OR CONTRACTOR CAN PROVIDE CONTINUOUSLY WELDED AND LOCKING–TYPE LONGITUDINAL JOINTS AND SEAMS ON DUCTS OPERATING AT STATIC PRESSURES LESS THAN 2 INCHES WATER GAUGE (W.G.) (500 PA) PRESSURE CLASSIFICATION.

B. FLEXIBLE DUCT CONNECTION;

1. CONTRACTOR SHALL BRACE DUCTWORK (AS REQUIRED) AT ALL FLEXIBLE CONNECTIONS TO ENSURE THAT DUCTWORK IS KEPT IN ALIGNMENT.

2. FLEXIBLE DUCT: PROVIDE INSULATED UL LISTED CLASS 1 DUCT COMPLYING WITH NFPA 90A, FLEXMATER, THERMAFLEX, WIREMOLD OR CLEVAFLEX. MAXIMUM LENGTH SHALL BE 8 FEET.

C. INSULATION

1. AIR CONDITIONING DUCTS REPAIR OR PROVIDE NEW INSULATION TO ALL OUTSIDE AIR SUPPLY AIR AND RETURN AIR DUCTS WITH A 1–1/2" THICK, 3/4 POUND DENSITY FIBERGLASS, FLEXIBLE BLANKET, INSULATION, FACED WITH A FIRE RESISTIVE VAPOR BARRIER JACKET WITH A 2" TAB ON ONE EDGE. INSULATION SHALL BE WRAPPED ON DUCTS WITH FACING OVERLAPPING ALL JOINTS AT LEAST 2" AND HELD IN PLACE WITH 1/2" OUTWARD CLINCHING STAPLES ON 4" CENTERS. STAPLES AND SEAMS ARE TO BE SEALED WITH A BRUSH COAT OF VAPOR BARRIER MASTIC.

2. INSULATION OF DUCTWORK AND PIPING PASSING THROUGH NO–RATED WALLS SHALL BE CONTINUOUS THROUGH THE WALL PENETRATION. DUCT AND PLENUM INSULATION AND SEALING: ALL SUPPLY AND 3. RETURN AIR DUCTS & PLENUMS SHALL BE INSULATED WITH A MINIMUM OF R–6 INSULATION WHERE LOCATED IN UNCONDITIONED SPACES & A MINIMUM OF R–8 INSULATION WHERE LOCATED OUTSIDE THE BUILDING. WHERE LOCATED WITHIN A BUILDING ENVELOPE ASSEMBLY, THE DUCT OR PLENUM SHALL BE SEPARATED FROM THE BUILDING EXTERIOR OR UNCONDITIONED OR EXEMPT SPACES BY A MINIMUM OF R–8 INSULATION.

4. DUCT TIGHTNESS SHALL BE VERIFIED BY EITHER OF THE FOLLOWING:

a. POST CONSTRUCTION TEST: TOTAL LEAKAGE SHALL BE LESS THAN OR EQUAL TO 4 CFM (113.3 L/MIN) PER 100 SQUARE FEET (9.29 M2) OF CONDITIONED FLOOR AREA WHEN TESTED AT A PRESSURE DIFFERENTIAL OF 0.1 INCHES W.G. (25 PA) ACROSS THE ENTIRE SYSTEM, INCLUDING THE MANUFAC–TURER'S AIR HANDLER ENCLOSURE. ALL REGISTER BOOTS SHALL BE TAPED OR OTHERWISE SEALED DURING THE TEST.

b. ROUGH–IN TEST: TOTAL LEAKAGE SHALL BE LESS THAN OR EQUAL TO 4 CFM (113.3 L/MIN) PER 100 SQUARE FEET (9.29 M2) OF CONDITIONED FLOOR AREA WHEN TESTED AT A PRESSURE DIFFERENTIAL OF 0.1 INCHES W.G. (25 PA) ACROSS THE SYSTEM, INCLUDING THE MANUFACTURER'S AIR HANDLER ENCLOSURE. ALL REGISTERS SHALL BE TAPED OR OTHERWISE SEALED DURING THE TEST. IF THE AIR HANDLER IS NOT INSTALLED AT THE TIME OF THE TEST, TOTAL LEAKAGE SHALL BE LESS THAN OR EQUAL TO 3 CFM (85 L/MIN) PER 100 SQUARE FEET (9.29 M 2) OF CONDITIONED FLOOR AREA.

c. THE TOTAL LEAKAGE TEST IS NOT REQUIRED FOR EXCEPTION DUCTS AND AIR HANDLERS LOCATED ENTIRELY WITHIN THE BUILDING THERMAL ENVELOPE.

d. A WRITTEN REPORT OF THE RESULTS OF THE TEST SHALL BE SIGNED BY THE PARTY CONDUCTING THE TEST AND PROVIDED TO THE CODE OFFICIAL UPON REQUEST.

e. AIR OUTLETS: ALL DIFFUSERS, REGISTERS AND GRILLES SHALL BE THE SIZES AND TYPE SHOWN AND AS MANUFACTURED BY METALAIRE OR APPROVED EQUAL AND SHALL HAVE FRAMES SUITABLE FOR THE SURFACES THEY ARE INSTALLED IN AND OF COLORS SELECTED BY THE ARCHITECT.

D. VIBRATION ISOLATORS:

1. PROVIDE NEOPRENE ISOLATION PADS FOR ALL EQUIPMENT 100 POUNDS OR GREATER.

2. QUANTITY AND LOCATION OF ISOLATORS SHALL BE AS RECOMMENDED BY THE EQUIPMENT MANUFACTURER.

E. REFRIGERANT PIPING:

1. INSULATE REFRIGERANT LINES TO A MINIMUM OF R–3. THE LIQUID LINE DOES NOT REQUIRE INSULATION.

2. PROTECTION OF PIPING INSULATION: PIPING INSULATION EXPOSED TO WEATHER SHALL BE PROTECTED FROM DAMAGE, INCLUDING THAT CAUSED BY SUNLIGHT, MOISTURE, EQUIPMENT MAINTENANCE, AND WIND, AND SHALL PROVIDE SHIELDING FROM SOLAR RADIATION THAT CAN CAUSE DEGRADATION OF THE MATERIAL. ADHESIVE TAPE SHALL NOT BE PERMITTED. WRAP WITH ALUMINUM AND SEAL WITH MASTIC, ALTERNATIVELY USE PLASTIC OR PVC SHIELDING.

23A.3 EXECUTION

A. PERFORM ALL OPERATIONS REQUIRED AND INSTALL ALL DUCTWORK, EQUIPMENT AND CONTROLS WITH ALL REQUIRED ACCESSORIES TO PRODUCE A COMPLETE HVAC INSTALLATION READY FOR USE.

1. INSTALL ALL DUCTWORK ABOVE CEILING AND HOLD TIGHT TO UNDERSIDE OF STRUCTURE ABOVE UNLESS OTHERWISE INDICATED.

2. ALL RETURN AIR DUCT OPENINGS ABOVE CEILING SHALL BE COVERED WITH 1/2" MESH SCREEN.

3. CHANGES TO DUCT DUE TO FIELD CONDITIONS SHALL BE MADE ONLY IF THE DUCT SIZE FREE AREA IS MAINTAINED AND SHALL BE SUBMITTED TO ENGINEER FOR APPROVAL.

B. TESTS: BEFORE ACCEPTANCE OF HVAC SYSTEMS, EACH SHALL BE TESTED, ADJUSTED AND BALANCED. THE AIR SYSTEM SHALL BE BALANCED TO DELIVER SPECIFIED AIR QUANTITIES WITHIN SPECIFIED CFM FOR EACH OUTLET OR INLET. EQUIPMENT AND CONTROLS ARE TO BE TESTED AND DETERMINED THAT ALL OPERATIONS AND SEQUENCES ARE SATISFACTORY.

1. ALL EQUIPMENT, DUCTWORK CONTROLS AND PIPING SHALL BE PROTECTED DURING THE COURSE OF CONSTRUCTION. ANY DAMAGED EQUIPMENT SHALL BE REPLACED AT NO EXTRA COST TO THE OWNER.

2. ALL WASTE AND DEBRIS WHICH IS PART OF THE HVAC WORK SHALL BE REMOVED FROM THE JOB SITE AT THE END OF EACH WORKING DAY. ALL EQUIPMENT AND DEVICES SHALL BE LEFT CLEAN AND IN WORKING ORDER.

MECHANICAL NOTES:

1. PRIOR TO BID, THE MECHANICAL CONTRACTOR SHALL EXAMINE ALL PROJECT DRAWINGS TO DEVELOP A COMPLETE UNDERSTANDING OF THE PROJECT SCOPE. FAILURE TO REVIEW ALL CONTRACT DOCUMENTS WILL NOT RELIEVE HIM OF THE RESPONSIBILITY TO PERFORM ALL WORK REQUIRED. UPON REVIEW OF ALL DOCUMENTS, CONTRACTOR SHALL ADVISE THE OWNER'S REPRESENTATIVE OF ANY DISCREPANCIES WHICH WILL AFFECT THE WORK REQUIRED.

2. CONTRACTOR IS RESPONSIBLE TO PROVIDE A COMPLETE AIR DISTRIBUTION SYSTEM (S) AS REQUIRED BY SMACNA STANDARDS AND A COMPLETE INSTALLATION OF ALL EQUIPMENT AS REQUIRED BY THE CODE AND THE UNIT MANUFACTURER AT NO ADDITIONAL COST TO THE OWNER REGARDLESS WHETHER THE ITEMS ARE SHOWN ON THE PLAN OR COVERED IN THE SPECIFICATIONS.

3. ALL NEW DUCTWORK SHALL RUN AS HIGH AS POSSIBLE ABOVE CEILING UNLESS NOTED OTHERWISE. ALL DUCT SIZES SHOWN ARE CLEAN INSIDE DIMENSIONS. 90 DEGREE ELBOW DUCTS ARE TO HAVE RADII OF 1.5 TIMES THE DUCT WIDTH. ALL BRANCH DUCTWORK CONNECTIONS TO MAIN AND DIFFUSER RUN–OUTS SHALL INCLUDE VOLUME DAMPERS.

4. FOR EXACT LOCATION OF ALL AIR DEVICES, SEE REFLECTED CEILING PLAN ON ARCHITECTURAL DRAWINGS.

5. OUTDOOR AIR SHALL HAVE BACKDRAFT DAMPER & MEET THE MAXIMUM LEAKAGE RATES.

6. ALL DUCT JOINTS, SEAMS, AND CONNECTIONS ARE TO BE SEALED TO SMACNA CLASS A REGARDLESS OF PRESSURE CLASS.

7. THE APPROVAL AND INSTALLATION OF FUEL GAS DISTRIBUTION PIPING AND EQUIPMENT, FUEL GAS APPLIANCES AND FUEL GAS FIRED APPLIANCE VENTING SYSTEMS SHALL BE IN ACCORDANCE WITH INTERNATIONAL FUEL GAS CODE AND MANUFACTURER'S REQUIREMENTS.

MECHANICAL NEW WORK NOTES:

A. EXHAUST CFM OF THE RANGE MICROWAVE EXHAUST HOOD SHALL BE NOT MORE THAN 400 CFM.

B. THIS PROJECT IS REQUIRED TO SUBMIT AT FINAL INSPECTION A DUCT LEAKAGE TEST SHOWING A PASSING RATING OF <= 4 CFM PER 100 SQUARE FEET CONDITIONED FLOOR AREA AT A PRESSURE OF 25 PASCAL. A WRITTEN REPORT OF THE RESULTS OF THE TEST SHALL BE SIGNED BY THE PARTY CONDUCTING THE TEST AND PROVIDED TO THE CODE OFFICIAL.

C. THIS PROJECT IS REQUIRED TO SUBMIT AT FINAL INSPECTION A WHOLE–BUILDING BLOWER DOOR TEST SHOWING A PASSING RATING OF <= 3 AIR CHANGES PER HOUR AT A PRESSURE OF 50 PASCAL. A WRITTEN REPORT OF THE RESULTS OF THE TEST SHALL BE SIGNED BY THE PARTY CONDUCTING THE TEST AND PROVIDED TO THE CODE OFFICIAL. TESTING SHALL BE CONDUCTED IN ACCORDANCE WITH ASTM E 779 OR ASTM E 1827.

D. HVAC PIPING CARRYING FLUIDS > 105 F OR FLUIDS <55 F ARE INSULATED WITH MINIMUM R–3 PIPE INSULATION. PIPING INSULATION EXPOSED TO WEATHER SHALL BE PROTECTED FROM DAMAGE, INCLUDING THAT CAUSED BY SUNLIGHT, MOISTURE, EQUIPMENT MAINTENANCE, AND WIND, AND SHALL PROVIDE SHIELDING FROM SOLAR RADIATION THAT CAN CAUSE DEGRADATION OF THE MATERIAL. ADHESIVE TAPE SHALL NOT BE PERMITTED.

E. ALL ENVIRONMENTAL AIR EXHAUST CAPS ON SIDE EXTERIOR WALL SHALL BE MINIMUM 3 FEET AWAY FROM PROPERTY LINES; 3 FEET FROM OPERABLE OPENINGS INTO BUILDINGS AND 10 FEET FROM MECHANICAL AIR INTAKES.

F. BUILDING CAVITIES SHALL NOT BE USED AS DUCTS OR PLENUMS.

G. AIR HANDLER LEAKAGE DESIGNED BY MANUFACTURER SHALL BE AT <2% OF AIRFLOW.

H. CONTRACTOR SHALL PROVIDE AUXILIARY DRAIN PAN UNDER EACH FAN COIL WITH A WATER–LEVEL DETECTION DEVICE TO AUTO SHUTDOWN UNIT WHEN CONDENSATE OVERFLOW OCCURS.

AIR DEVICES SCHEDULE

CFM RANGE	LISTED SIZE	FACE SIZE	NC MAX.	MODEL	REMARKS
0–75	2 X 12	3–1/8 X 13–1/5	30	LIMA MODEL 875	REGISTER IN KICK PLATE UNDER SINK
0–50	6X6	7–1/2 X 7–1/2	30	LIMA SERIES 600	WALL AND CEILING DIFFUSER
51–100	8X6	9–1/2 X 9–1/2	30	LIMA SERIES 600	WALL AND CEILING DIFFUSER
101–150	12X6	13–1/2 X 7–1/2	30	LIMA SERIES 600	WALL AND CEILING DIFFUSER
151–205	14X8	15–1/2 X 9–1/2	30	LIMA SERIES 600	WALL AND CEILING DIFFUSER
0–125	2–1/2 X 12	4–7/8 X 13–7/8	30	LIMA SERIES 667	FLOOR DIFFUSER
126–250	4 X 14	6–7/8 X 15–7/8	30	LIMA SERIES 667	FLOOR DIFFUSER
0–600	10 X 10	12 X 12	30	LIMA STEEL / RETURN FILTERED GRILLE 60G	WALL/CEILING GRILLE
0–1500	22 X 22	24 X 24	30	LIMA STEEL / RETURN FILTERED GRILLE 60G	WALL/CEILING GRILLE

NOTES:
① ALL CEILING DIFFUSER TO BE PROVIDED WITH:
A. SQUARE TO ROUND NECK TRANSITION.
B. OPPOSED BLADE DAMPER.
② SEE ARCH. REFLECTED CEILING PLAN FOR DIFFUSERAND REGISTER FOR EXACT LOCATION.

WASHINGTON DC CODES:

BUILDING:	2015 IBC AND 2017 DCMAR12A SUPPLEMENT 2015 IEBC
COVERING CODES:	2017 DISTRICT OF COLUMBIA BUILDING CODE
MECHANICAL:	2017 DISTRICT OF COLUMBIA MECHANICAL CODE
ELECTRICAL:	2014 INTERNATIONAL NFPA–70
PLUMBING:	2017 DISTRICT OF COLUMBIA PLUMBING CODE
FIRE:	2017 DISTRICT OF COLUMBIA FIRE CODE
GREEN:	2017 DISTRICT OF COLUMBIA GREEN CONSTRUCTION CODE
ENERGY:	2017 DISTRICT OF COLUMBIA ENERGY CONSTRUCTION CODE

MECHANICAL SYMBOLS

- FLEXIBLE DUCTWORK (BUILDING STANDARD)
- NEW DUCTWORK (WIDTH X DEPTH INSIDE CLEAR DIMENSION)
- NEW DUCT W/1" SOUND LINING
- VD VOLUME DAMPER
- FIRE DAMPER
- RETURN AIR GRILLE OPEN TO CEILING SPACE (BLDG STANDARD)
- (T) PROGRAMMABLE THERMOSTAT
- (U) DOOR UNDER–CUT 3/4" MIN.
- RETURN/EXHAUST AIR
- REMOVED
- CONNECT TO EXISTING
- POINT OF REMOVAL
- (M) MOTORIZED VOLUME DAMPER

ABBREVIATIONS

CD	CEILING DIFFUSER
COND	CONDENSATE
CFM	CUBIC FEET PER MINUTE
CLG	CEILING
CR	CEILING REGISTER
DB	DRY BULB
DN	DOWN
(E)	EXISTING
EA	EXHAUST AIR
EF	EXHAUST FAN
ER	EXISTING RELOCATED
ESP	EXTERNAL STATIC PRESSURE
ETR	EXISTING TO RELOCATE
GAL	GALLONS
(N)	NEW
OA	OUTSIDE AIR
OBD	OPPOSED BLADE DAMPER
RA	RETURN AIR
R	REMOVE
RR	REMOVE AND RELOCATE
SP	STATIC PRESSURE
WR	WALL REGISTER
T'STAT	THERMOSTAT
(TYP)	TYPICAL
V	VENT
VD	VOLUME DAMPER
WMS	WIRE MESH SCREEN

SHEET INDEX:

M100	MECHANICAL COVER SHEET AND SCHEDULES
M200	MECHANICAL FLOOR PLANS
M300	MECHANICAL DETAILS

RESIDENTIAL VENTILATION REQUIREMENTS:

RESIDENTIAL VENTILATION IS BASED ON NUMBER OF ROOMS.

	REQUIRED	PROVIDED
FOR 3 BEDROOM (AHU–1): →	60 CFM O.A.	60 CFM O.A.

ENERGY NOTES

A. SUPPLY AND RETURN DUCTS OUTSIDE OF THE BUILDING THERMAL ENVELOPE SHALL BE INSULATED TO A MINIMUM OF R–8.

B. ALL JOINTS AND SEAMS OF AIR DUCTS, AIR–HANDLERS, AND FILTER BOXES ARE SEALED.

C. AIR HANDLER LEAKAGE DESIGNED BY MANUFACTURER SHALL BE AT ≤ 2% OF AIRFLOW. IN ACCORDANCE WITH ASHRAE 193.

D. TOTAL DUCT LEAKAGE TEST ≤4 CFM/100 FT² WITH AIR–HANDLER INSTALLED.

E. BUILDING CAVITIES SHALL NOT BE USED AS DUCTS OR PLENUMS.

F. MECHANICAL SYSTEM PIPING CAPABLE OF CARRYING FLUIDS ABOVE 105°F (41°C) OR BELOW 55°F (13°C) SHALL BE INSULATED TO A MINIMUM OF R–3.

G. PIPING INSULATION EXPOSED TO WEATHER SHALL BE PROTECTED FROM DAMAGE, INCLUDING THAT CAUSED BY SUNLIGHT, MOISTURE, EQUIPMENT MAINTENANCE AND WIND, AND SHALL PROVIDE SHIELDING FROM SOLAR RADIATION THAT CAN DEGRADATION OF THE MATERIAL. ADHESIVE TAPE SHALL NOT E BE PERMITTED.

H. HVAC PIPING INSULATION, WHERE EXPOSED TO THE OUTDOORS, SHALL BE FINISHED WITH TWO COATS OF MANUFACTURER'S FINISH COATING, VINYL–LACQUER COATING OR APPROVED EQUAL.

I. CONTRACTOR SHALL PROVIDE MOTORIZED DAMPER ON OUTSIDE AIR INTAKE DUCTWORK. DAMPER SHALL BE INTERLOCKED WITH THE AIR HANDLING UNIT. DAMPER SHALL BE OPEN WHEN THE AHU IS OPERATING, AND CLOSE WHEN THE AHU IS NOT OPERATING.

J. THERMOSTAT SHALL INCLUDE THE CAPABILITY TO SET BACK OR TEMPORARILY OPERATE THE SYSTEM TO MAINTAIN ZONE TEMPERATURES DOWN TO 55°F (13°C) OR UP TO 85°F (29°C). THE THERMOSTAT SHALL INITIALLY BE PROGRAMMED BY THE MANUFACTURER WITH A HEATING TEMPERATURE SET POINT NO HIGHER THAN 70°F (21°C) AND A COOLING TEMPERATURE SET POINT NO LOWER THAN 78°F (26°C).

K. ALL THE CEILING TYPE BATHROOM EXHAUST FANS SHALL BE PROVIDED WITH GRAVITY BACKDRAFT DAMPERS.

L. NEW EQUIPMENT AND APPLIANCES SHALL BE SEALED COMBUSTION.

M. PROJECT IS REQUIRED TO SUBMIT AT FINAL INSPECTION A WHOLE–BUILDING BLOWER DOOR TEST SHOWING A PASSING RATING OF <= 3 AIR CHANGES PER HOUR AT A PRESSURE OF 50 PASCAL PER R402.4.1.2. A WRITTEN REPORT OF THE RESULTS OF THE TEST SHALL BE SIGNED BY THE PARTY CONDUCTING THE TEST AND PROVIDED TO THE CODE OFFICIAL. TESTING SHALL BE CONDUCTED IN ACCORDANCE WITH ASTM E 779 OR ASTM E 1827.

N. FOR ALL NEW FUEL BURNING APPLIANCES AND EQUIPMENT LOCATED INSIDE THE BUILDING ENVELOPE MUST BE SEALED COMBUSTION PER R402.4.4.

SPLIT SYSTEM HEAT PUMP UNIT SCHEDULE

SYSTEM	TONNAGE	SUPPLY AIR	OUTSIDE AIR	SENSIBLE COOLING (MBH) @ 95 AMB	EAT (DB/WB)	LAT (DB/WB)	TOTAL COOLING (MBH)	HEATING TYPE	HEATING CAPACITY MBH(OUTPUT)	REFRIGERANT TYPE	SEER	BASIS OF DESIGN INDOOR UNIT	OUTDOOR UNIT
AHU–C01 & COND–C01	5.0 TON	2000 CFM	60 CFM	45.0 MBH	80 F/67.0 F	57 F/55 F	60 MBH	HEAT PUMP / ELECTRIC HEAT 15 KW	61.0 MBH	R–410A	16.0	CARRIER FB4CNF060 240V/1PH/60HZ 43.0 MCA/70.0 MOCP 30.0 LBS.	CARRIER 25HCE460 240V/1PH/60HZ 21.5 MCA/30.0 MOCP 212 LBS.

NOTES:

1. ALL COOLING CAPACITIES ARE BASED ON 80°F DB, 63°F WB INDOOR ENTERING AIR TEMP AND 95°F AMBIENT OUTDOOR ENTERING AIR TEMP, 45°F SUCTION TEMP.
2. PROVIDE SYSTEMS WITH PROGRAMMABLE THERMOSTATS. TEMPERATURE SET POINT HEATING AT 70°F AND COOLING AT 78°F. AUX. HEAT TEMP. MUST DISPLAY ON THE SCREEN. WHEN THE TEMPERATURE RANGE FALLS BELOW 35°F (ADJ.) THE AUXILIARY HEAT TURNS ON.
3. ESP IS EXCLUSIVE OF FILTERS, WET COIL, AND CASING LOSS.
4. HEATING AND COOLING VALVES ARE MINIMUM REQUIRED TO MEET DESIGN.
5. FURNISH UNIT WITH LOW AMBIENT CONTROLS.
6. AIR HANDLERS SHALL HAVE A MANUFACTURER'S DESIGNATION FOR AN AIR LEAKAGE OF NO MORE THAN 2 PERCENT OF THE DESIGN AIR FLOW RATE WHEN TESTED IN ACCORDANCE WITH ASHRAE 193. REFER TO SUBMITTED DOCUMENT FROM UNIT MANUFACTURER.
7. UNITS SHALL MEET ENERGY STAR.
8. THE ELECTRIC RESISTANCE SHALL TURN ON, ONLY WHEN THE HEAT PUMP CAN'T HANDLE THE LOAD. THE AUX. HEAT MODE IS NORMAL WHEN: THE TEMPERATURE OUTSIDE IS BELOW FREEZING AND HEAT PUMP IN DEFROST MODE.

FAN SCHEDULE

MARK	LOCATION	CFM	ESP IN WG.	ELECTRICAL DATA					DRIVE	FAN CONTROLLED BY	MANUFACTURER AND MODEL NUMBER	REMARKS
				VOLTS	PHASE	HERTZ	RPM	WATTS				
EF–1 THRU 6	SEE PLAN	75	0.125	120	1	60	900	54	DIRECT	WALL SWITCH	GREENHECK SP–80–VG	SEE NOTES BELOW

NOTE:
1. ALL FANS SHALL BE PROVIDED WITH BACKDRAFT DAMPER AND INTEGRAL DISCONNECT SWITCH.
2. ALL TOILET EXHAUST FANS SHALL BE CONTROLLED BY WALL SWITCH AND ENERGY STAR.

5105 LEE STREET NE
DISTRICT OF COLUMBIA
20019

ONE DESIGN SERVICES
organizationfornewenvironment@gmail.com
Tel. 571.225.7211

DISTRICT OF COLUMBIA
KHALID LAMAAFI
PE922123
PROFESSIONAL ENGINEER
12/29/2022

ENGINEER: KHALID LAMAAFI
ADDRESS: 9707 ASPEN PL.
MANASSAS VA, 20110
EMAIL: ALLATMEP@GMAIL.COM
CELL: 571-833-8357

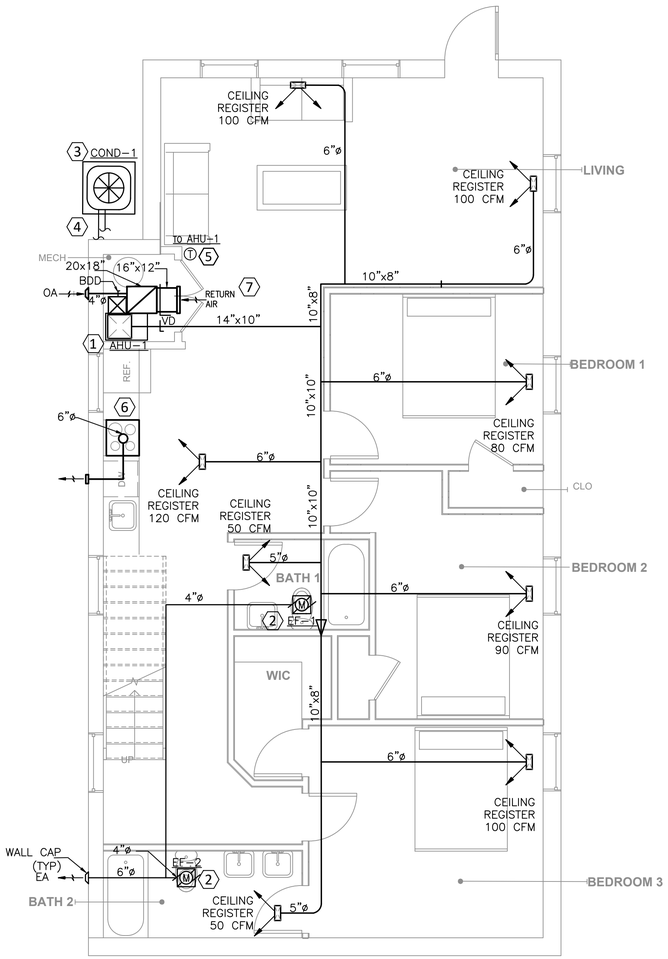
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SCALE	AS INDICATED
ISSUE DATE	DEC 2022

MECHANICAL COVER SHEET AND SCHEDULES

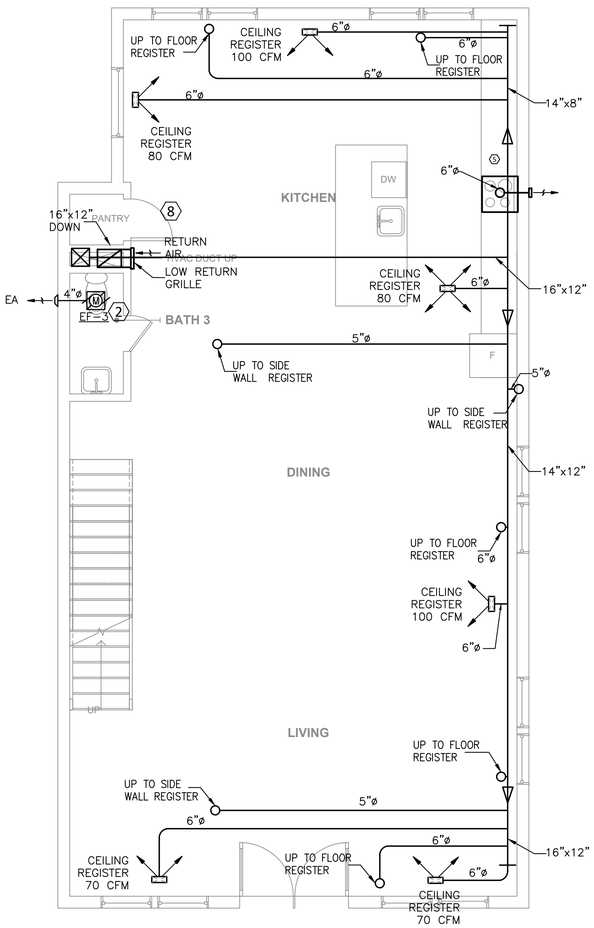
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⟨-⟩ MECHANICAL KEYED NOTES:

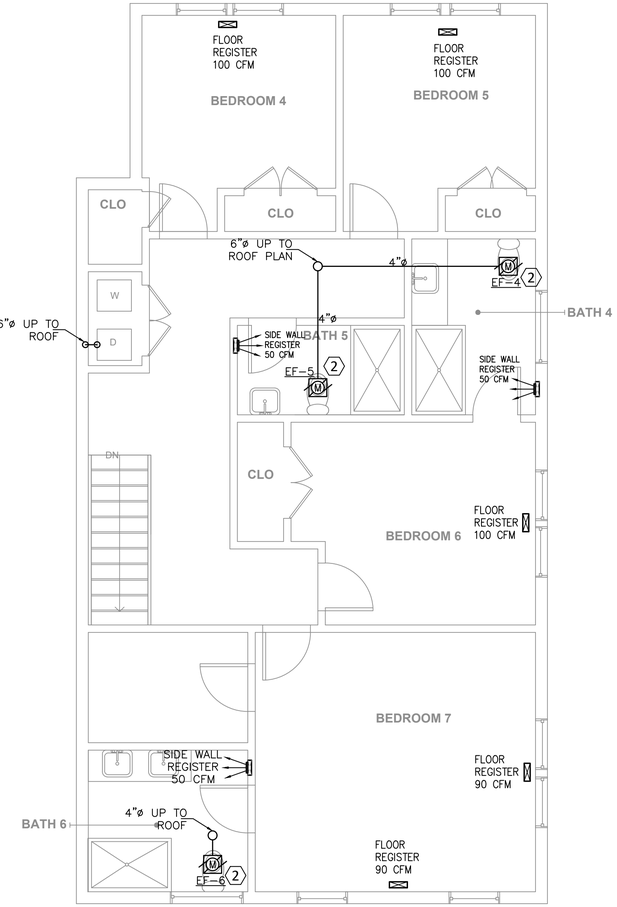
1. PROVIDE NEW AHU AT THIS LOCATION. REFER TO SCHEDULE AND DETAIL FOR MORE INFORMATION. ROUTE 3/4"CONDENSATE DRAIN TO DAYLIGHT. PROVIDE PUMP DRAIN IF NEEDED.
2. PROVIDE BATHROOM EXHAUST FAN. REFER TO SCHEDULE AND DETAILS. INSTALL AS PER MANUFACTURER'S INSTRUCTIONS.
3. PROVIDE NEW CONDENSING UNIT. REFER TO SCHEDULE AND DETAIL FOR MORE INFORMATION. INSTALL AS PER MANUFACTURER'S INSTRUCTIONS.
4. ROUTE REFRIGERANT SUCTION & LIQUID PIPING FROM INDOOR AHU TO OUTDOOR UNIT. SIZING TO BE BASED ON MANUFACTURER'S REQUIREMENTS FOR TOTAL DEVELOPED LENGTH.
5. PROVIDE NEW THERMOSTAT TO CONTROL AHU. COORDINATE EXACT LOCATION WITH OWNER/ARCH.
6. DUCTED HOOD.
7. FILTERED RETURN AIR GRILLE.



1 MECHANICAL CELLAR FLOOR PLAN
SCALE: 1/4"=1'-0"



2 MECHANICAL FIRST FLOOR PLAN
SCALE: 1/4"=1'-0"



3 MECHANICAL SECOND FLOOR PLAN
SCALE: 1/4"=1'-0"

5105 LEE STREET NE
DISTRICT OF COLUMBIA
20019

ONE DESIGN SERVICES
organizationfornewenvironment@gmail.com
Tel. 571.225.7211

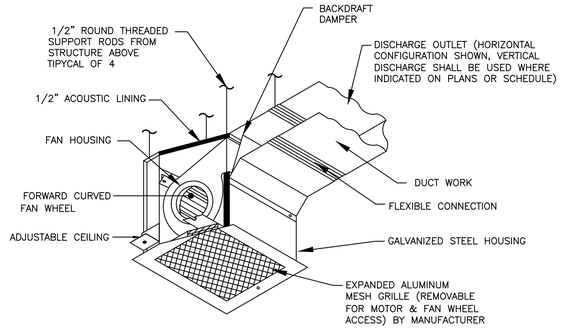
DISTRICT OF COLUMBIA · KHALID LAMAAFI · PE922123 · PROFESSIONAL ENGINEER
12/29/2022

ENGINEER: KHALID LAMAAFI
ADDRESS: 9707 ASPEN PL.
MANASSAS VA, 20110
EMAIL: ALLATMEP@GMAIL.COM
CELL: 571-833-8357

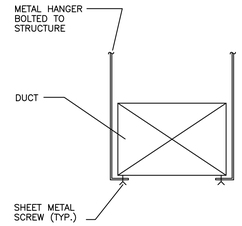
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SCALE	AS INDICATED
ISSUE DATE	DEC 2022

MECHANICAL FLOOR PLAN

M200



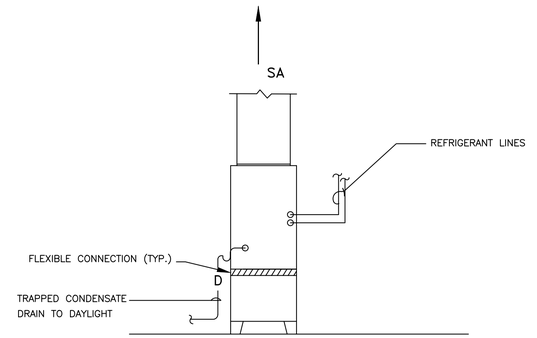
1 CABINET CEILING EXHAUST FAN DETAIL
N.T.S.



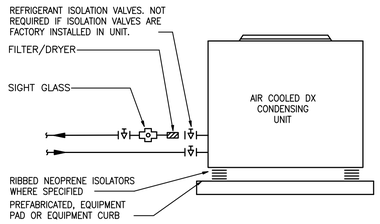
NOTES:

1. ALL DUCTWORK AND EQUIPMENT SUPPORT TO BUILDING STRUCTURE SHALL COMPLY WITH THE LATEST EDITION OF THE SMACNA STANDARDS FOR UPPER ATTACHMENT DEVICES.
2. CONTRACTOR SHALL COORDINATE TYPE HANGERS AND ANCHORING SYSTEM WITH THE BUILDING STRUCTURE.

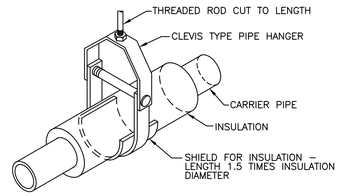
2 TYPICAL DUCT SUPPORT DETAILS
N.T.S.



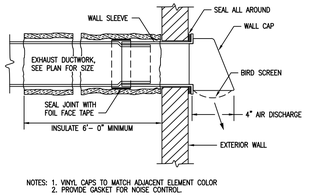
3 VERTICAL AHU INSTALLATION DETAIL
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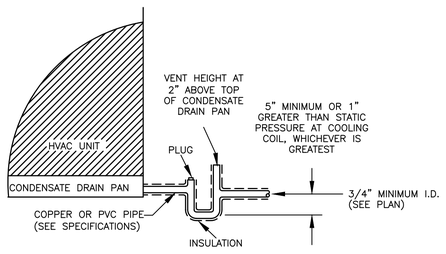
4 TYPICAL CONDENSING UNIT PIPING DETAIL
N.T.S.



5 CLEVIS TYPE PIPE HANGER FOR INSULATED PIPE
N.T.S.



6 WALL CAP DETAIL
N.T.S.



NOTE: ROUTE DRAIN TO STORM DRAIN AND DAYLIGHT

7 CONDENSATE DRAIN DETAIL
N.T.S.

5105 LEE STREET NE
DISTRICT OF COLUMBIA
20019

ONE DESIGN SERVICES
organizationfornewenvironment@gmail.com
Tel. 571.225.7211

DISTRICT OF COLUMBIA KHALID LAMAAFI PE922123 PROFESSIONAL ENGINEER
12/29/2022

ENGINEER: KHALID LAMAAFI
ADDRESS: 9707 ASPEN PL.
MANASSAS VA, 20110
EMAIL: ALLATMEP@GMAIL.COM
CELL: 571-833-8357

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MECHANICAL DETAILS

M300

PLUMBING SPECIFICATIONS

GENERAL

A. ALL WORK SHALL CONFORM TO THE REQUIREMENTS OF ALL CODES, ORDINANCES AND STANDARDS OF THE LOCAL JURISDICTION.IN CASE OF A CONFLICT BETWEEN DRAWINGS OR SPECIFICATIONS AND THE REQUIREMENTS OF THE LOCAL JURISDICTION, THE MORE STRINGENT REQUIREMENTS SHALL APPLY.

B. ALL WORK SHALL BE GUARANTEED AGAINST DEFECTS, LEAKS, LACK OF PROPER SYSTEM PERFORMANCE OR NON-OPERATION FOR A PERIOD OF ONE YEAR AFTER DATE OF ACCEPTANCE.

C. ALL WORK SHALL BE COORDINATED WITH ALL TRADES, PRIOR TO INSTALLATION.

D. IN GENERAL, DRAWINGS FOR THE WORK ARE DIAGRAMMATIC AND SHOW THE LOCATION, TYPE AND SIZE OF PIPING, EQUIPMENT, AND ACCESSORY EQUIPMENT. THE CONTRACTOR SHALL FURNISH ALL ITEMS NECESSARY FOR THE PROPER INSTALLATION AND OPERATION OF THE WORK, WHETHER CALLED FOR OR NOT. THE CONTRACTOR SHALL VERIFY ALL NECESSARY DIMENSIONS BEFORE INSTALLING ANY OF THE WORK AND SHALL CHECK HIS LAYOUTS TO ALLOW CLEARANCE REQUIRED FOR OTHER WORK. THE SCOPE OF WORK CONSISTS GENERALLY OF PROVIDING AND INSTALLING COMPLETE PLUMBING AND GAS SYSTEMS AND FINAL TESTING OF ALL SYSTEMS AND EQUIPMENT AS REQUIRED.

PRODUCTS

A. PLUMBING FIXTURES: ALL FIXTURES SHALL BE SELECTED BY OWNER. PROVIDE ALL FIXTURES WITH TRIM, CARRIER SUPPLIES, AND TRAPS AS REQUIRED FOR COMPLETE INSTALLATION.

B. PIPING AND FITTING:

1. DOMESTIC WATER:
ABOVE GRADE SHALL BE TYPE "L" HARD DRAWN COPPER TUBING WITH 125 PSI WROUGHT COPPER SWEAT FITTINGS, AND ALL JOINT SOLDERED WITH 95/5 OR SILVER SOLDER. BELOW GRADE SHALL BE TYPE "K" SOFT DRAWN COPPER TUBING WITH 125 PSI WROUGHT COPPER SWEAT FITTINGS SOLDERED WITH SILVER SOLDER.

2. SOIL, WASTE AND VENT:
ABOVE GRADE SHALL BE: SERVICE WEIGHT CAST IRON BELL AND SPIGOT.SCHEDULE 40 GALVANIZED STEEL PIPE WITH SWEAT. CAST IRON DRAINAGE PATTERN FITTINGS. CAST IRON NO-HUB PIPING AND FITTINGS.DWV COPPER TUBING AND COPPER DRAINAGE PATTERN FITTINGS.SCHEDULE 40 PVC PLASTIC PIPE AND PVC-DWV FITTING. (SHALL NOT BE USED IN PLENUM SPACES.) BELOW GRADE SHALL BE: SERVICE WEIGHT CAST IRON BELL AND SPIGOT. SOIL, WASTE AND VENT STACKS SHALL BE SERVICE WEIGHT CAST IRON BELL AND SPIGOT.

C. INSULATION:

1. DOMESTIC WATER PIPING:
COVER ALL WITH 1/2" FIBERGLASS INSULATION (R3 SECURED WITH ALL PURPOSE JACKET. PIPING IN EXTERIOR WALLS AND PLUMBING CHASES SHALL BE COVERED WITH 1" THICK INSULATION.

2. STORM WATER PIPING:
THE HORIZONTAL SECTION OF THE RAIN LEADERS, RISER TO AND INCLUDING THE INTERIOR PART OF THE ROOF DRAIN SHALL BE COVERED WITH 1" THICK INSULATION.

D. VALVES:

1. DOMESTIC WATER:
ALL VALVES SHALL BE SWEATED BRONZE GATE VALVE WITH SCREW-IN BONNET, RISING STEM MINIMUM RATING OF 125 PSI. TWO PIECES BALL VALVES WITH EXTENDED HANDLE MAY BE USED IN LIEU OF THE GATE VALVES.

E. HANGERS:
SHALL BE ADJUSTABLE CLEVIS HANGERS, PROPERLY SIZED AND SPACED FOR PIPING, INCLUDING INSULATION.

EXECUTION

A. INSTALL FIXTURES LEVEL, PLUMB AND PARALLEL TO WALLS. ALL EXPOSED METAL PARTS SHALL BE CHROME PLATED AND SHOW NO TOOL MARKS. GROUT BETWEEN WALL HUNG FIXTURES AND WALL. PROVIDE ACCESS PANELS TO ALL CONCEALED SUPPLY STOPS AND TRAP.

B. FIXTURES DESIGNATED FOR USE B PHYSICALLY HANDICAPPED PEOPLE SHALL BE IN ACCORDANCE WITH ANSI A 117.1.

C. INSTALL DIELECTRIC CONNECTION BETWEEN DISSIMILAR METALS, PIPE TO PIPE, PIPE TO EQUIPMENT, PIPE TO SUPPORT.

5. FURNISH AND INSTALL JOSAM 75000 SERIES SHOCK ARRESTERS AT THE ENDS OF ALL HOT AND COLD WATER BRANCHES TO FIXTURES. SIZES SHALL BE IN ACCORDANCE WITH PLUMBING AND DRAINAGE INSTITUTE STANDARD P.D.1

PLUMBING NOTES:

1. ALL WORK SHALL BE DONE IN ACCORDANCE WITH THE STATE BUILDING CODE, LOCAL REQUIREMENTS AND THE PROJECT SPECIFICATIONS.
2. ALL WORK SHALL BE COORDINATED WITH ALL OTHER TRADES.
3. ALL DRAINAGE PIPING SHALL BE RUN AT A SLOPE OF 1/4" PER FOOT UNLESS SPECIFICALLY SHOWN OTHERWISE ON DRAWINGS.
4. CONTRACTOR SHALL VISIT SITE PRIOR TO SUBMISSION OF BID TO BECOME FAMILIAR WITH EXISTING CONDITIONS.
5. ALL HOT AND COLD WATER SUPPLY PIPING SHALL BE INSULATED.
6. PROVIDE AND INSTALL CLEAN-OUTS IN DRAINAGE PIPING AT EACH CHANGE IN DIRECTION OF PIPING GREATER THAN 45 DEGREES, EVERY 50 FEET, AND AS SHOWN.
7. EXPOSED UTILITY SERVICE LINES AND PIPES SHALL BE INSTALLED SO THAT THEY DO NOT OBSTRUCT OR PREVENT CLEANING OF THE FLOORS, WALLS, OR CEILINGS. EXPOSED HORIZONTAL UTILITY SERVICE LINES AND PIPES SHALL NOT BE INSTALLED ON THE FLOOR.
8. CONTRACTOR TO VERIFY SIZE AND LOCATION OF SANITARY, AND COLD/HOT WATER PIPES PRIOR TO STARTING WORK.
9. EXISTING UTILITIES AND EQUIPMENT NOT SHOWN OR NOT SHOWN TO BE REPLACED SHALL REMAIN IN SERVICE DURING CONSTRUCTION.
10. CONTRACTOR SHALL REMOVE AND DISPOSE ALL PLUMBING MATERIAL, FIXTURES AND EQUIPMENT FROM TENANT SPACE AS SHOWN ON DRAWING. COORDINATE DEMOLITION WITH NEW CONSTRUCTION PLAN.
11. NEW OR REPAIRED POTABLE WATER SYSTEMS SHALL BE PURGED OF DELETERIOUS MATTER AND DISINFECTED PRIOR TO UTILIZATION. THE METHOD TO BE FOLLOWED SHALL BE THAT PRESCRIBED BY THE HEALTH AUTHORITY OR WATER PURVEYOR HAVING JURISDICTION OR, IN THE ABSENCE OF A PRESCRIBED METHOD, THE PROCEDURE DESCRIBED IN EITHER AWWA C651 OR AWWA C652, OR AS DESCRIBED IN 2017 DCPC SECTION 610. THIS REQUIREMENT SHALL APPLY TO "ON-SITE" OR "IN-PLANT" FABRICATION OF A SYSTEM OR TO A MODULAR PORTION OF A SYSTEM.

PLUMBING LEGEND

———	SANITARY PIPE
——G———	GAS PIPE
— — — — —	VENT PIPE
——— - ———	DOMESTIC COLD WATER PIPE
——— - - ———	DOMESTIC HOT WATER PIPE
—|—	CLEAN OUT
	PIPE UP, PIPE DOWN
	SHUT-OFF GATE VALVE, VALVE IN VERTICAL
—||—	PIPE UNION
	FLOOR DRAIN

PLUMBING ABBREVIATIONS

CO	CLEANOUT
CW	COLD WATER
DN	DOWN
FD	FLOOR DRAIN
GAL	GALLONS
GPH	GALLONS PER HOUR
GPM	GALLONS PER MINUTE
HB	HOSE BIBB
HW	HOT WATER
LAV	LAVATORY
PSI	POUND PER SQUARE INCH
OSD	OPEN SITE DRAIN
SAN	SANITARY
TYP	TYPICAL
V	VENT
VTR	VENT THRU ROOF
WC	WATER CLOSET

WASHINGTON DC CODES:

BUILDING:	2015 IBC AND 2017 DCMAR12A SUPPLEMENT 2015 IEBC
COVERING CODES:	2017 DISTRICT OF COLUMBIA BUILDING CODE
MECHANICAL:	2017 DISTRICT OF COLUMBIA MECHANICAL CODE
ELECTRICAL:	2014 INTERNATIONAL NFPA-70
PLUMBING:	2017 DISTRICT OF COLUMBIA PLUMBING CODE
FIRE:	2017 DISTRICT OF COLUMBIA FIRE CODE
GREEN:	2017 DISTRICT OF COLUMBIA GREEN CONSTRUCTION CODE
ENERGY:	2017 DISTRICT OF COLUMBIA ENERGY CONSTRUCTION CODE

SHEET INDEX:

P100	PLUMBING COVER SHEET
P200	PLUMBING DETAILS
P300	PLUMBING SANITARY FLOOR PLANS
P400	PLUMBING DOMESTIC WATER FLOOR PLANS
P500	PLUMBING RISER DIAGRAMS

PLUMBING FIXTURE CONNECTION SCHEDULE

MARK	FIXTURE	CONNECTIONS					REMARKS
		COLD WATER	HOT WATER	WASTE	VENT	MAX. FLOW RATE	
WC	WATER CLOSET	1/2''	-	3"	2"	1.28 GPF	PROVIDE BY OWNER
LAV	LAVATORY	1/2"	1/2"	2''	2"	1.5 GPM	AMERICAN STANDARD LUCERNE 20-1/2" WALL MOUNTED PORCELAIN BATHROOM SINK MODEL:0356.041.020, GENTA SINGLE HOLE SINGLE-HANDLE BATHROOM FAUCET IN CHROME, MODEL #WS84760.
KS	KITCHEN SINK	1/2"	1/2"	2"	1-1/2"	1.8 GPM	STAINLESS STEEL, DOUBLE BOWL KITCHEN SINK, 1.5 GPM FAUCET.
FD	FLOOR DRAIN	-	-	2''	2"	-	WATTS MODEL FD-100-A FLOOR DRAINS.
REF	REFRIGERATOR	1/2"	-	-	-	-	PROVIDE BY OWNER
BT	BATH TUBE	1/2"	1/2"	2"	1-1/2"	-	PROVIDE BY OWNER
CO	CLEANOUT	-	-	2"	-	-	ZURN MODEL: ZN 1400-3NH (SQUARE POLISHED NICKEL.)

NOTE:

1. COORDINATE WITH ARCH./OWNER PRIOR TO PURCHASE.
2. SET TEMPERING VALVE AT 105' F. VALVES SHALL MEET ASSE 1070.
3. WATER CLOSET SHALL BE TANK TYPE WITH HINGED OPEN FRONT SEAT, SIZED FOR BOWL TYPE.
4. TRAPS INSTALLED IN FLOOR/CEILING (TUB AND SHOWER) AND WALL (CLOTHES WASHING MACHINE)
5. SPACES WON'T BE OF THE SLIP JOIT TYPE.
6. COMPLY WITH ANSI A117,1 FOR ACCESSIBLE FIXTURE'S MOUNTING HEIGHTS.
7. SHOWER COMPARTMENT AND TUB/SHOWER WALL SURFACES MUST BE CONSTRUCTED OF SMOOTH,
8. CORROSION-RESISTANT AND NONABSORBENT WATERPROOF MATERIALS.

PLUMBING EQUIPMENT SCHEDULE:

EQUIPMENT	DESCRIPTIONS
MV1, THERMOSTATIC MIXING VALVES.	WATTS MMV SERIES THERMOSTATIC MIXING VALVE, MODEL MMV-US-M1
HB, HOSE BIB	ZURN MODEL Z1345, EXPOSED NON-FREEZE WALL FAUCET WITH EXTERNAL VACUUM BREAKER
GD, GARBAGE DISPOSER	INSINKERATOR FOOD WASTE DISPOSER MODEL SS-100-5-MRS, 1 HP, 115V/1PH/60HZ, 5.1 AMP

WATER HEATER SCHEDULE - (EWH)

MARK	AREA SERVED	RECOVERY AT 90° F			CAPACITY (GALLONS)	ELECTRIC DATA		REMARKS
		GPH	EWT	LWT		VOLT/PH/HZ	KW	
EWH	SEE PLAN	21	40	120	55	240/01/60	4.5 KW	BRADFORD, MODEL: RE255T6

(N) = NEW

(E) = EXISTING

5105 LEE STREET NE
DISTRICT OF COLUMBIA
20019

ONE DESIGN SERVICES
organizationfornewenvironment@gmail.com
Tel. 571.225.7211

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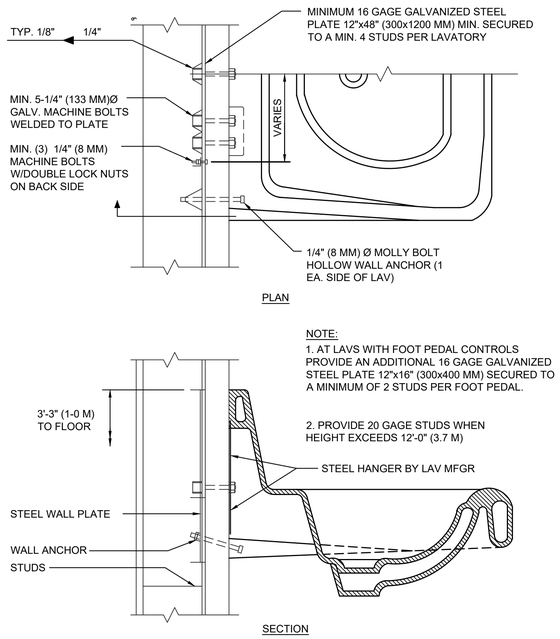
12/29/2022

ENGINEER: KHALID LAMAAFI
ADDRESS: 9707 ASPEN PL.
MANASSAS VA, 20110
EMAIL: ALLATMEP@GMAIL.COM
CELL: 571-833-8357

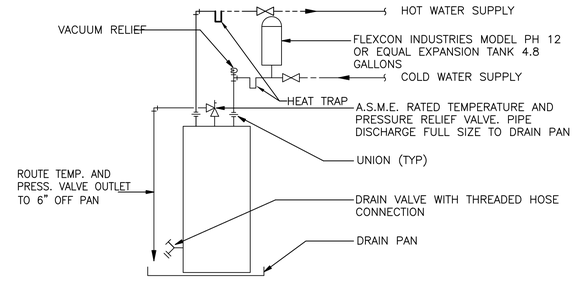
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PLUMBING COVER SHEET

P100



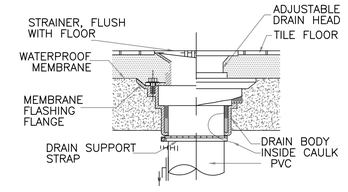
1 LAVATORY MOUNTING DETAIL
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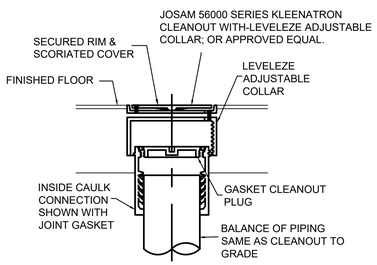
NOTES:

1. PROVIDE 0.5" MINIMUM INSULATION FOR DOMESTIC COLD WATER PIPES.
2. PROVIDE 1" MINIMUM INSULATION FOR DOMESTIC HOT WATER PIPES (MIN.3-R).

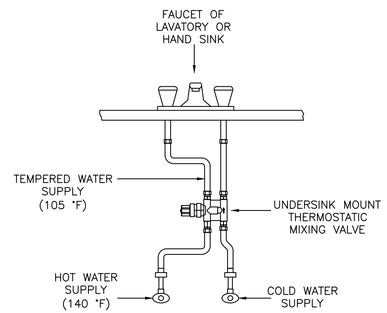
2 ELECTRIC WATER HEATER PIPING DETAIL
N.T.S.



3 FLOOR DRAIN
N.T.S.



4 FLOOR CLEANOUT DETAIL
N.T.S.



5 UNDERSINK MOUNT THERMOSTATIC
N.T.S.

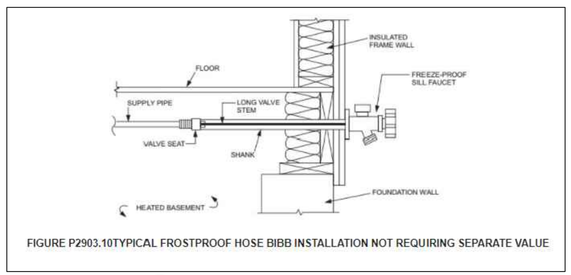


FIGURE P2903.10TYPICAL FROSTPROOF HOSE BIBB INSTALLATION NOT REQUIRING SEPARATE VALUE

6 HOSE BIBB DETAIL
N.T.S.

5105 LEE STREET NE
DISTRICT OF COLUMBIA
20019

ONE DESIGN SERVICES

organizationfornewenvironment
@gmail.com
Tel. 571.225.7211

DISTRICT OF COLUMBIA KHALID LAMAAFI PE922123 PROFESSIONAL ENGINEER
12/29/2022

ENGINEER: KHALID LAMAAFI
ADDRESS: 9707 ASPEN PL.
MANASSAS VA, 20110
EMAIL: ALLATMEP@GMAIL.COM
CELL: 571-833-8357

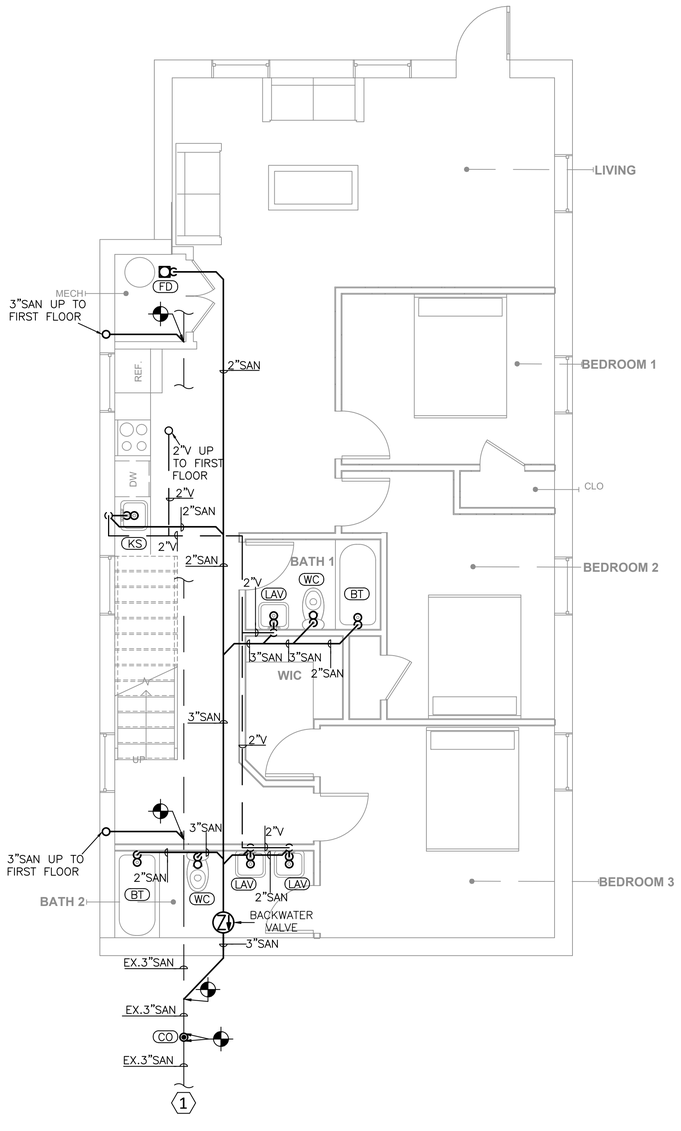
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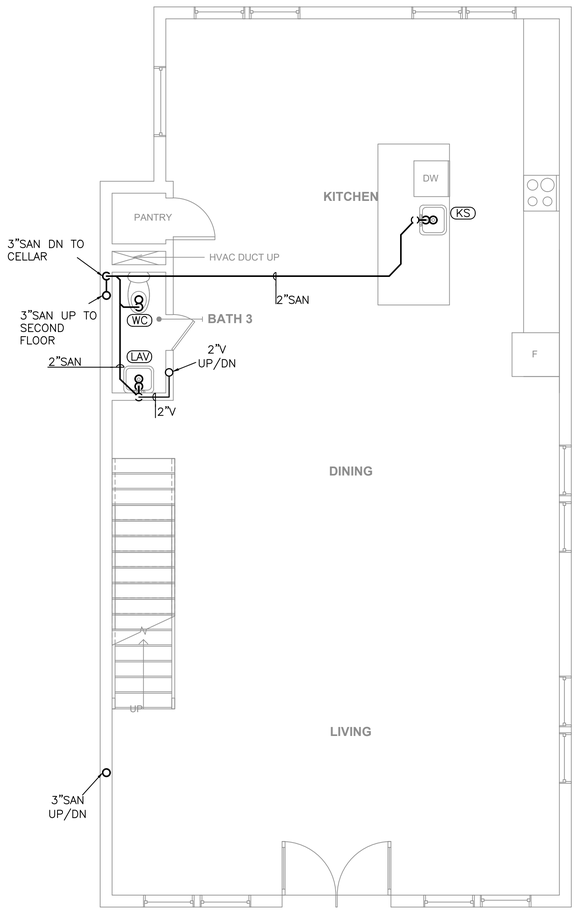
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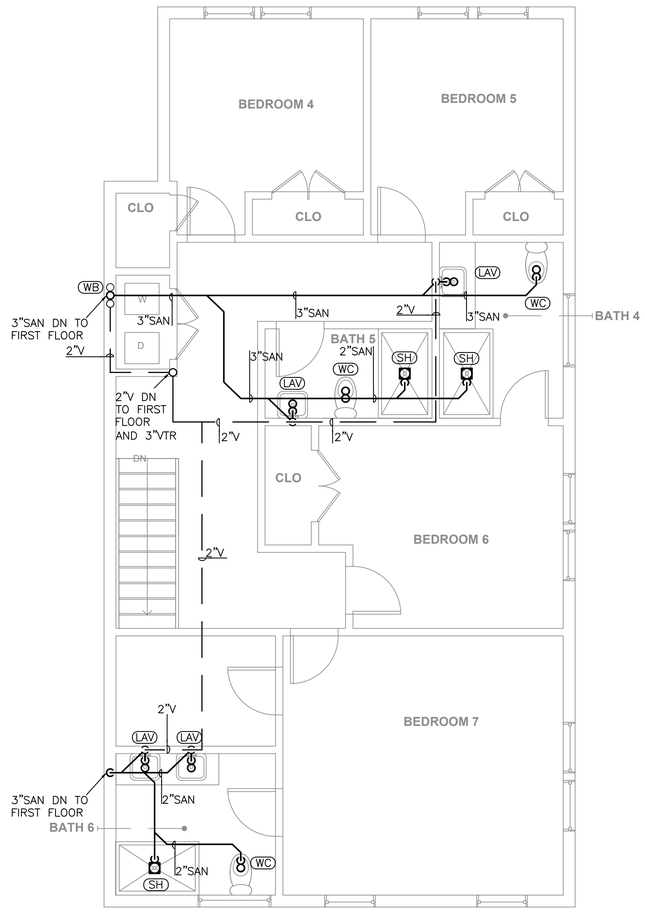
1. CONTRACTOR TO VERIFY EXACT LOCATION OF EXISTING PIPING IN THE FIELD.



1 PLUMBING SANITARY CELLAR FLOOR PLAN
SCALE: 1/4"=1'-0"



2 PLUMBING SANITARY FIRST FLOOR PLAN
SCALE: 1/4"=1'-0"



3 PLUMBING SANITARY SECOND FLOOR PLAN
SCALE: 1/4"=1'-0"

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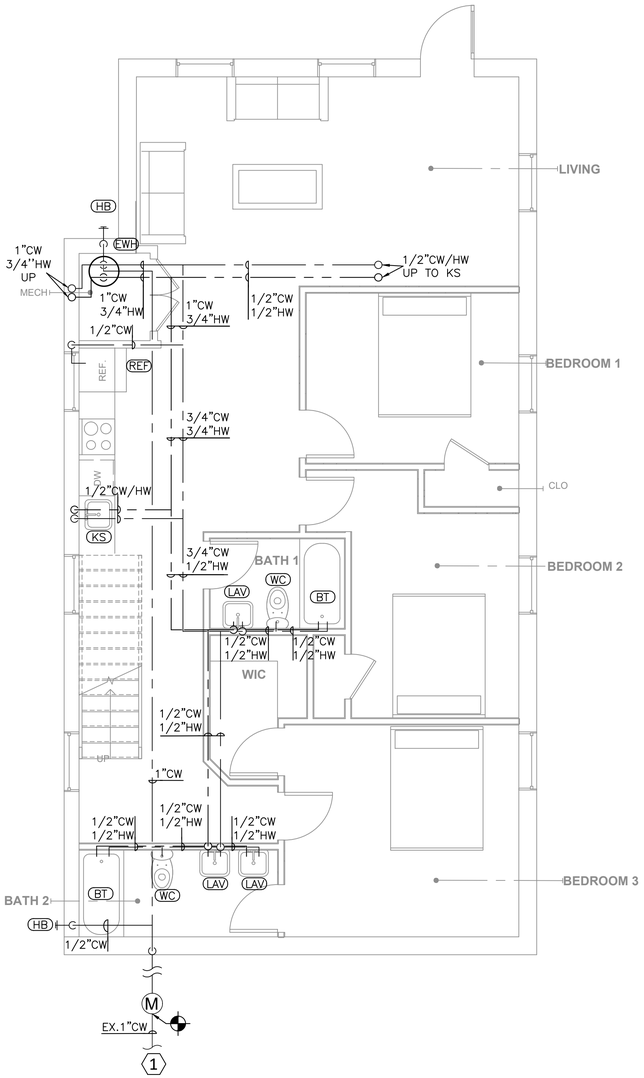
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PLUMBING SANITARY FLOOR PLANS

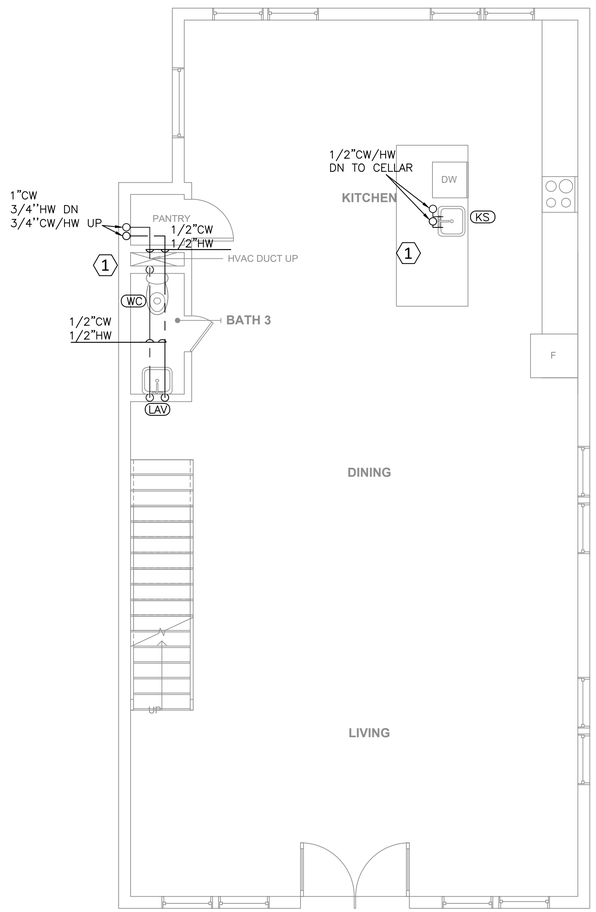
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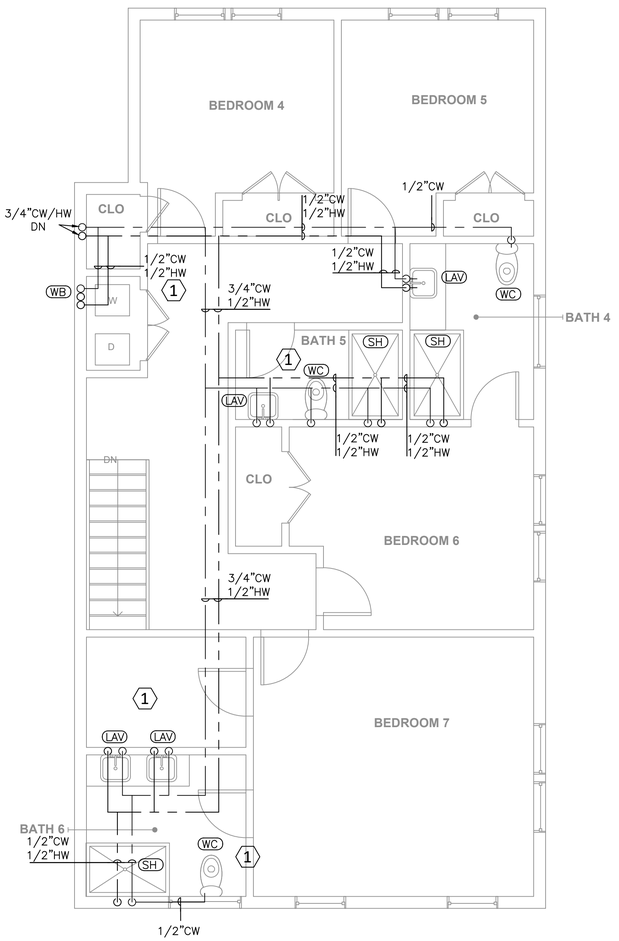
(1) CONTRACTOR TO VERIFY EXACT LOCATION OF EXISTING PIPING IN THE FIELD.



1 PLUMBING DOMESTIC WATER CELLAR FLOOR PLAN
SCALE: 1/4"=1'-0"



2 PLUMBING DOMESTIC WATER FIRST FLOOR PLAN
SCALE: 1/4"=1'-0"



3 PLUMBING DOMESTIC WATER SECOND FLOOR PLAN
SCALE: 1/4"=1'-0"

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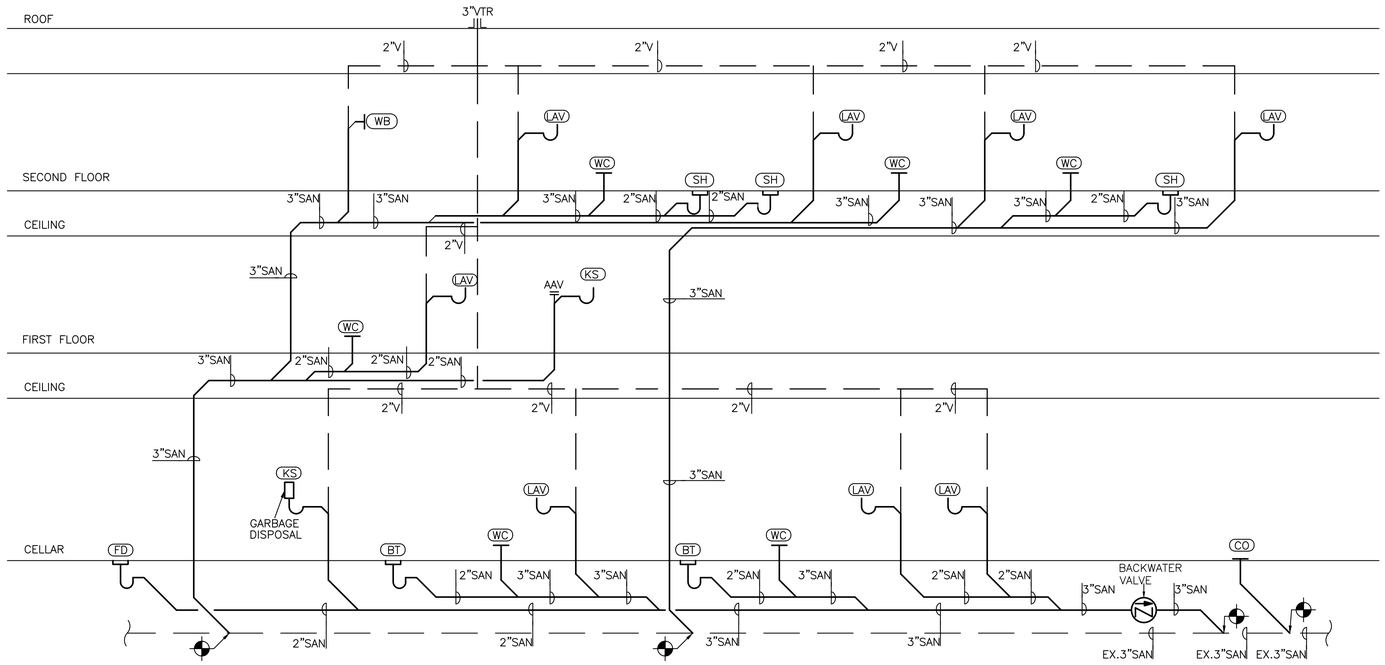
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MANASSAS VA, 20110
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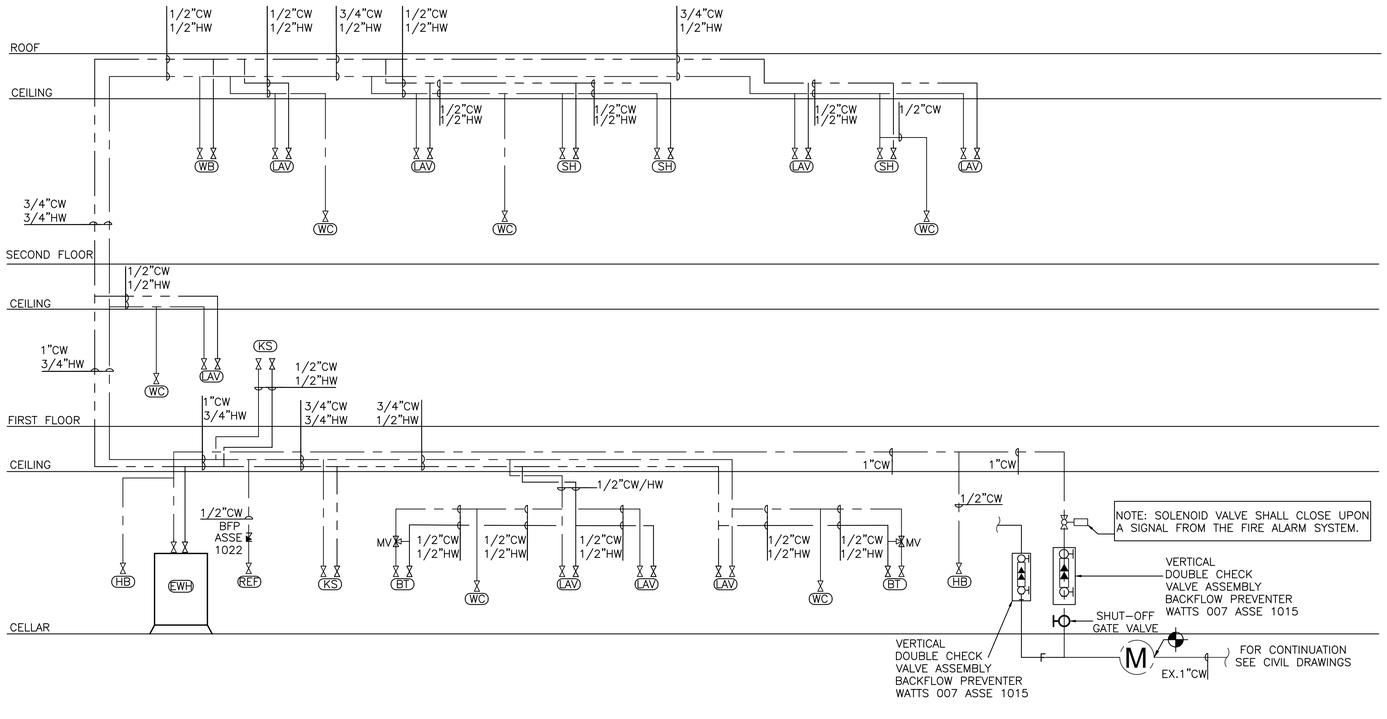
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PLUMBING DOMESTIC WATER FLOOR PLANS

P400



1 PLUMBING SANITARY RISER DIAGRAM
N.T.S.



2 PLUMBING DOMESTIC WATER RISER DIAGRAM
N.T.S.

5105 LEE STREET NE
DISTRICT OF COLUMBIA
20019

ONE DESIGN SERVICES
organizationfornewenvironment@gmail.com
Tel. 571.225.7211

DISTRICT OF COLUMBIA KHALID KAMAL LAMAAFI PE922123 PROFESSIONAL ENGINEER
12/29/2022

ENGINEER: KHALID LAMAAFI
ADDRESS: 9707 ASPEN PL.
MANASSAS VA, 20110
EMAIL: ALLATMEP@GMAIL.COM
CELL: 571-833-8357

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PLUMBING RISER DIAGRAMS

P500